

ENERGY-VALUE SUPERVISION FOR DE NOVO MOLECULAR FLOWS: LOCAL DENSITY ORDERING WITHOUT PARTITION FUNCTIONS

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ABSTRACT

De novo 3D molecular generation methods learn distributions over composition and geometry, while energy-targeted samplers assume a fixed system or a supplied molecular graph. We introduce *energy-value supervision* for the tractable positional density of a bond-free de novo flow: flow matching plus one group-wise objective, penalising the variance of $\log q_\theta(x \mid c) + E(c, x)/kT$ over local perturbations of a parent geometry, where q_θ is read out by reversing the positional ODE with the discrete labels clamped. The partition function cancels within each group, and the same divergence can be estimated off-policy under an arbitrary reference measure, so the energy labels come from a transferable universal potential and are pre-computed once, with no bond labels and no model rollouts. On a conservative held-out evaluation of 93 disjoint parents scored by an independent GFN2-xTB potential, the method raises the mean within-parent density-energy correlation from 0.200 to 0.369, top-1 retrieval of the lowest-energy geometry from 20.2% to 29.6% and top-3 from 49.5% to 62.4%. Across completed runs, a matched six-cell study shows the gain requires the correct geometry-energy correspondence: zero-energy and scrambled-energy controls do not improve over flow matching. The gain persists on fixed-RMSD shells, ruling out radial-distance variation as the sole explanation. An identifiability analysis characterises the objective’s scope and identifies connected supervision as the missing ingredient for controlling relative basin mass, and raw calibration does not improve, indicating that ranking, calibration and global allocation ask for distinct interventions.

1 INTRODUCTION

De novo 3D molecular generators create composition and geometry at once (Hoogeboom et al., 2022; Xu et al., 2023; Irwin et al., 2025; Daigavane et al., 2024; Dunn & Koes, 2025; Morehead et al., 2026): they propose which atoms a molecule contains and where those atoms sit, which a conformer generator handed a graph does not do. What they lack is a way to make the geometries they prefer the geometries a physical potential prefers, and that ingredient now exists. Universal machine-learned potentials return an energy $E(c, x)$ of near-DFT quality for arbitrary compositions at negligible cost (Levine et al., 2025; Fu et al., 2025), transferring across chemical space rather than being fitted per system. Energy-targeted samplers already train tractable densities against $e^{-E/kT}$ (Noé et al., 2019; Klein & Noé, 2025; Tan et al., 2025; Rehman et al., 2026; Havens et al., 2025; Akhound-Sadegh et al., 2024; Dern et al., 2025), but they assume a fixed system, or a supplied composition or graph, so their machinery does not reach a generator that is still choosing what molecule to make. The opportunity is to use energy *values* to shape the density a de novo flow already defines; Section 2.1 places both literatures against that question, and Appendix C extends the discussion.

We do this with a single additional term. A continuous normalising flow assigns a tractable log-density to any geometry by reversing its own positional ODE and accumulating the divergence (Grathwohl et al., 2019); we call this the *clamped positional density* $\log q_\theta(x \mid c)$, the discrete channels being held at their endpoint labels while the positions evolve. Over K displaced geometries of one parent, all sharing a composition c , we penalise $\text{Var}_k[\log q_\theta(x_k \mid c) + E(c, x_k)/kT]$, which is zero exactly when the density is proportional to $e^{-E/kT}$ on that group (Section 2, Figure 1). Centring inside a group deletes the unknown normaliser $Z(c)$. Separately, the same divergence can be estimated under

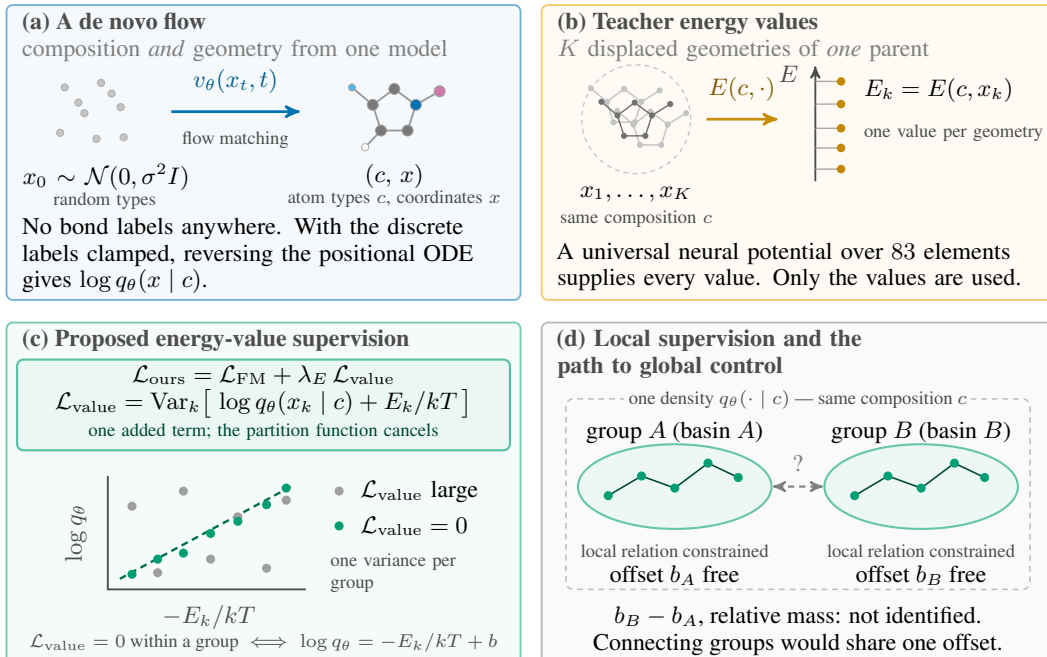


Figure 1: **Energy-value supervision without a partition function.** Schematic; no measured quantity appears here. (a) With the endpoint discrete labels clamped, reversing the positional ODE defines the *clamped positional-flow density* $\log q_\theta(x | c)$, which need not equal the joint generator’s conditional density. (b) A universal neural potential values each of K displaced geometries of one parent. (c) The proposed objective is flow matching plus the within-group variance of $\log q_\theta(x_k | c) + E_k/kT$, which vanishes exactly when $\log q_\theta = -E_k/kT + b$ inside the group: the partition function cancels and the offset b is free. (d) Two disconnected groups of the *same* density keep independent offsets, so the term constrains the local density–energy relation inside each group and leaves their relative mass unidentified; connecting groups so they share one offset is the extension toward global control. Panel detail in Appendix B.18.

an arbitrary reference measure, and that is what makes the supervision *off-policy*: the objective is an expectation under whatever measure produced the displaced geometries, not under the model, so every energy label is computed once, ahead of training, and read from disk like data: one extra loss, pre-computed labels, no bond supervision, no rollouts, no reference sampler.

The intervention works, and we show what in it works. On 93 held-out parents that the supervision pool never contained, with every geometry re-scored by a GFN2-xTB potential that enters no training loss, the mean per-parent correlation between $\log q_\theta$ and $-E/kT$ rises from 0.200 ± 0.018 over five flow-matching seeds to 0.369 ± 0.023 over four completed value seeds, a gain of $+0.169$ in which all four value seeds exceed all five control seeds. It converts into a decision: ranking a parent’s displaced geometries by $\log q_\theta$ retrieves the lowest-energy one 29.6% of the time against 20.2%, with 12.5% at chance. A matched six-cell grid then localises the effect, since zeroed energies and energies scrambled within a parent leave ordering where flow matching leaves it, so what helps is the correct geometry-to-energy correspondence and not the presence of an auxiliary penalty; a control holding displacement magnitude fixed keeps the gain, ruling out radial-distance variation as the sole explanation. Finally an identifiability analysis says how far the term reaches: it fixes the density–energy relation inside a group up to one additive constant per connected component of the group-overlap graph, making group connectivity a design variable and relative basin mass something a further objective must supply (Section 3).

- **Method.** Energy-value supervision for de novo molecular flows: a group-wise variance objective, whose divergence form is standard and is credited in Section 2, applied to the clamped positional density of a bond-free generator over an 83-element corpus, with pre-computed values from a transferable potential, no bond labels, no rollouts and no partition function.
- **Controlled evidence.** A matched six-cell study and a conservative held-out endpoint under an independent scorer, showing improved local ordering and retrieval and attributing both to

the correct geometry–energy pairing rather than to regularisation, the energy multiset alone, or displacement magnitude.

- **Characterisation and design guidance.** A result identifying what group-wise value supervision determines — the local density–energy relation up to one offset per connected component of the overlap graph — turning the objective’s scope into a design criterion: local groups suffice for local ordering, and connected groups or a global term are what basin mass requires (Section 3).

2 ENERGY-VALUE SUPERVISION FOR DE NOVO MOLECULAR FLOWS

2.1 RELATED WORK: TWO LITERATURES, AND THE QUESTION BETWEEN THEM

De novo 3D molecular generation. Equivariant diffusion and flow-matching models generate positions and types jointly, from EDM and GeoLDM (Hooeboom et al., 2022; Xu et al., 2023) through graph-and-geometry hybrids (Vignac et al., 2023; Daigavane et al., 2024; Irwin et al., 2025) to FlowMol3 (Dunn & Koes, 2025), the backbone used here, and Zatom-1 (Morehead et al., 2026), the one published generator trained on the same corpus; EBMol (Griesbacher et al., 2026) learns a scalar energy from data rather than from a physical potential. These models are judged on validity, on sample quality, or on an energy the model itself learns, and they define the state of the art at the task they set out to do. What such evaluations generally do not ask is whether the tractable density the model defines responds to an *external* physical energy. That is a different question rather than a shortfall — it is not the question these papers pose — and we claim superiority over none of them.

Conditional conformer and ensemble models. Torsional diffusion (Jing et al., 2022), the GEOM conformer literature (Axelrod & Gómez-Bombarelli, 2022) and DECAF (Moqvist et al., 2026) take the molecular graph, or the molecular identity, as input and return conformers for it. The composition is supplied rather than generated, so the density they define is conditional on a fixed chemical graph. That input is what makes theirs a different problem from a de novo generator’s, and why their protocols do not transfer to one unchanged.

Energy-targeted samplers, and where our objective comes from. Samplers that target $e^{-E/kT}$ directly (Noé et al., 2019; Klein & Noé, 2025; Tan et al., 2025; Rehman et al., 2026; Havens et al., 2025; Akhound-Sadeh et al., 2024; Dern et al., 2025) obtain equilibrium densities; mass-covering correction by annealed importance sampling (Midgley et al., 2023) addresses the cross-group failure that Corollary 2 formalises, potential-score matching (Guo et al., 2025) supervises with the potential’s gradient instead of its value, and Zhou et al. (2025) drive a predictor–refiner with an *internal, learned* energy instead of an external one. Each takes the system, the alphabet or the molecular graph as given: Adjoint Sampling (Havens et al., 2025), the closest use of a universal potential as a training signal, trains against an eSEN energy but samples conformers of a *given* molecular graph. They therefore solve a stronger sampling problem — equilibrium, not ordering — on a state space that is narrower and fixed in advance. Our value term is inherited from this literature and is not claimed as new: $\text{Var}[\log q_\theta + E/kT]$ is the *log-variance divergence* named by Richter et al. (2020), placed in a path-space family by Nüsken & Richter (2021), given an optimal-control reading by Berner et al. (2024) and established for diffusion samplers by Richter & Berner (2024); that a variance annihilates the normalising constant, and that the divergence may be estimated off-policy under an arbitrary reference measure (Sendra et al., 2024), are structural properties of that family, and we claim neither.

The intersection. Existing de novo generators decide what molecule and what geometry to produce, but do not test the tractable density they define against a transferable physical energy; energy-targeted samplers test or optimise exactly that relation, but assume the molecular system or the graph. This paper works at the intersection, asking what group-wise energy *values* from a universal potential do to the density of a generator that is still choosing its molecule. To our knowledge the combination studied here — an external universal potential supervising the tractable density of a bond-free de novo flow — has not been evaluated in this form. Appendix C gives the formal setting and the fuller discussion.

2.2 THE BOND-FREE FLOW AND ITS CLAMPED POSITIONAL DENSITY

A molecular state is a pair (c, x) : a composition c , the element multiset together with the total charge, and a centre-of-mass-free conformation $x \in \mathbb{R}^{3N}$. The backbone is FlowMol3 (Dunn & Koes, 2025), an SE(3)-equivariant conditional flow-matching model (Lipman et al., 2023; Liu et al., 2023; Albergo et al., 2023) with $x_t = (1 - t)x_0 + tx_1$, $x_0 \sim \mathcal{N}(0, \sigma^2 I)$ at $\sigma = 1$, and $\mathcal{L}_{\text{FM}}$ the velocity-regression loss; types and charges ride a continuous-time Markov chain. *De novo* describes that backbone and its task; every energy measurement here is instead conditional on a fixed endpoint composition, types and charges held at their data values whenever the teacher is queried. The teacher is the universal neural potential trained on OMol25 (Levine et al., 2025; Fu et al., 2025): 83 elements, DFT energies and forces, no bond labels, making the generator bond-free by necessity, bonds recovered post hoc where an evaluation needs them (Kim & Kim, 2015). It supplies $E(c, x)$ in eV and $F = -\nabla_x E$, so $\pi(x | c) = e^{-E(c, x)/kT} / Z(c)$; neither $Z(c)$ nor $\sum_c Z(c)$ is ever formed, and $kT = 1.0$ eV is a *numerical* target that keeps E/kT tame, not a physical temperature. Queries pass charge only and default to a singlet, so every evaluation below is on main-group organic parents (Assumption 6).

Because the generator is a continuous normalising flow, its log-density follows from the instantaneous change of variables (Grathwohl et al., 2019): integrating the field backwards from x_1 and accumulating the divergence gives

$$\log q_\theta(x_1 | c) = \log p_0(x_0) - \int_0^1 \nabla \cdot v_\theta(x_t, t) dt, \quad (1)$$

with the trace estimated by Rademacher–Hutchinson probes (Hutchinson, 1989). Throughout, $q_\theta(x | c)$ is the *clamped positional density*: (1) integrated with the discrete channels, atom types, formal charges and the empty bond tensor, held at their endpoint labels for the whole reverse trajectory, only the positions evolving. It costs one reverse solve per geometry, is normalised on the centre-of-mass-free subspace by construction, and is the object this paper supervises and evaluates. **It need not equal the joint generator’s conditional density** $p_\theta(x | c)$: the two coincide only if the discrete trajectory is determined by its endpoint, and we have not quantified the difference (Assumption 2).

2.3 THE GROUP-WISE VALUE OBJECTIVE

The intervention constrains the *value* of that log-density rather than its gradient. The Boltzmann residual $w(x) = \log q_\theta(x | c) + E(c, x)/kT$ is constant exactly where $q_\theta(\cdot | c) \propto e^{-E(c, \cdot)/kT}$, so asking only that w be *flat* across a set of geometries asks for the physics and nothing else. We pre-compute shards of K perturbations per parent, each carrying its teacher energy; every step draws M parents and penalises the variance of the residual inside each group:

$$\begin{aligned} \mathcal{L}_{\text{ours}} &= \mathcal{L}_{\text{FM}} + \lambda_E \mathcal{L}_{\text{value}}, & \mathcal{L}_{\text{value}} &= \frac{1}{M} \sum_{m=1}^M \text{Var}_k(w_{m,k}), \\ w_{m,k} &= \log q_\theta(x_{m,k} | c_m) + E_{m,k}/kT. \end{aligned} \quad (2)$$

where $w_{m,k}$ is the residual at the k -th geometry of parent m , evaluated with that parent’s own composition c_m and pre-computed teacher energy $E_{m,k}$. We use $\lambda_E = 3 \times 10^{-5}$ (code field: `lambda_2`), zero over a flow-matching warm-up and then ramped, applied every few steps because each evaluation is a reverse-time solve carrying second derivatives (Appendix B.2). Nothing else changes: same backbone, optimiser and data, no bond labels, and the model we report as ours sets $\lambda_1 = 0$ and the anchor weight $\lambda_3 = 0$.

2.4 WHY THE PARTITION FUNCTION CANCELS, AND WHY GROUPING IS MANDATORY

Three properties of (2) do the work. First, $\log Z(c)$ enters w as an additive constant and a variance annihilates additive constants, so the normaliser cancels within a group and is never formed — the reason values from a transferable potential are usable at all, since $Z(c)$ changes with every composition. Second, grouping is mandatory rather than convenient, precisely because that constant differs across molecules: its spread is set by size and composition rather than by deviation from the Boltzmann law, so a variance over a batch of *different* molecules would measure the spread of $\log Z$ instead (Remark 11). Third, and separately from the cancellation, (2) is a group-wise estimate,

against the unnormalised Gibbs weight, of the *log-variance divergence*, which may be estimated under an *arbitrary* reference measure. That, and not the cancellation, is what makes the supervision off-policy and pre-computed shards valid: no sample need come from the model. The divergence, the cancellation of $Z(c)$ inside it, and its off-policy estimation are prior work, credited in Section 2.1. Ours is the deployment: a group-conditioned use of that estimator inside a mixed discrete–continuous, bond-free de novo generator, at the price of one free constant per group (Section 3).

Practicalities. (i) λ_E was fixed once from a smoke run before any Boltzmann evaluation and never tuned against a reported metric; the weight is small, its *contribution* is not. (ii) $\log q_\theta$ comes from an ODE integration of the clamped positional flow: (1) advanced by explicit Euler steps over the whole of $[0, 1]$, its divergence integral taken by the midpoint rule on the same interval and its trace by Hutchinson probes (Hutchinson, 1989); the interval is not truncated, so the discretisation controls numerical error rather than which density is meant (Appendix B.2). (iii) The reverse-time solve back-propagates second derivatives through a stochastic trace, and runs carrying it reach non-finite weights at a rate that runs without it do not (6 of 20 against 0 of 10 in the matched grid of Section 4.1); a guard aborts on the first non-finite parameter, and because divergence is not independent of the objective, **every value-cell mean we report is conditional on completion**, with launched, completed and diverged counts beside it. (iv) The gradient through $\mathcal{L}_{\text{value}}$ is a *truncated-adjoint surrogate*: the reverse trajectory is advanced without gradient tracking, each state detached from the previous, so gradient flows only through the divergence evaluations; we do not differentiate through the solve and do not claim exact likelihood training (Appendix B.1).

2.5 A GRADIENT-SUPERVISION BASELINE

The obvious alternative to values is the potential’s gradient, measured separately under the same backbone and budget. Flow matching gives the score of its time- t marginal in closed form (Lemma 1), $s_\theta(x_t, t) = [t v_\theta(x_t, t) - x_t] / [(1 - t) \sigma^2]$, which at the Boltzmann optimum equals F/kT , so a natural comparator regresses one on the other,

$$\mathcal{L}_{\text{force}} = \mathbb{E}_{t^*, (c, x_1)} \left[N(c)^{-1} \sum_i \ell(s_\theta(x_{t^*}, t^*)_i - \frac{F(c, x_1)_i}{kT}) \right]. \quad (3)$$

The read-out has a pole at $t = 1$, so we probe only at $t^* \in \{0.85, 0.92, 0.97\}$ and average a per-atom Huberised, norm-capped error ℓ (Appendix B.2). The force target is read at the data endpoint x_1 while the score is read on the conditional path, so this is a naive off-policy endpoint form of force matching, a comparator and not a component of (2): the model we report as ours has $\lambda_1 = 0$, and results from $\lambda_1 > 0$ are labelled as such (Section 4.1).

3 FROM LOCAL ORDERING TO CONNECTED SUPERVISION

(2) sums *within-group* variances, so the question worth asking is what a model that drove it to zero would have been shown; the answer is precise enough to use as a design rule. It concerns q_θ rather than $p_\theta(x | c)$, and the zero set of the ideal residual condition rather than the implemented update’s trajectory. Write $w(x) = \log q_\theta(x | c) + E(c, x)/kT$, let ρ_j be the reference measure of group j with support U_j , and join two groups by an edge of the *overlap graph* $\mathcal{G}_c$ when ρ_j and $\rho_{j'}$ are not mutually singular, a property of the measures for which $U_j \cap U_{j'} \neq \emptyset$ is necessary but far from sufficient (Definition 1).

Proposition 1 (What zero group-wise variance fixes). *Let the population objective $\sum_j \pi_j \text{Var}_{\rho_j}[w]$ vanish. Then (i) $w \equiv b_j$ ρ_j -almost surely on U_j for a scalar b_j : the residual is fixed to one constant per group, and no normaliser is needed, since $Z(c)$ enters w only through that constant. (ii) $b_j = b_{j'}$ along every edge of $\mathcal{G}_c$, so the constants agree within each connected component; connectivity is sufficient and, in the identifiability sense, necessary, a disconnected graph leaving zero-loss solutions whose constants differ. (iii) If the ρ_j have positive Lebesgue densities and $\mathcal{G}_c$ has $C \geq 2$ components $U^{(1)}, \dots, U^{(C)}$ carrying all the model’s mass, the zero-loss densities form a $(C - 1)$ -parameter family $q_\theta = a_i e^{-E/kT}$ on $U^{(i)}$ with $\sum_i a_i Z^{(i)} = 1$, in which every vector of component occupancies in the open simplex is attained: the relative mass of separated components is not identified.*

Theorem 2 states this in full, with the connected case and its two further hypotheses, and Appendix A.6 proves it. Our runs sit in the empirical reading (Corollary 3): ρ_j is the empirical measure on a group’s

K sampled geometries, zero loss is $K - 1$ scalar equations at those points and nothing about unsampled ones, and parts (ii) and (iii) survive only as an identifiability statement.

What our design supervises. Each parent contributes exactly one group, and distinct parents draw displacements independently from continuous distributions, so with probability one no two groups share a sampled point: $\mathcal{G}_c$ is a set of isolated vertices and only part (i) applies (Corollary 2). Part (i) is exactly the property the experiments exercise — inside each sampled cloud the objective pins the relation between $\log q_\theta$ and energy, which is local ordering — and its complement is equally exact, the cloud’s offset being free and the relative probability of distinct basins untouched. Disconnection rests on mutual singularity rather than on one group per composition, which our pool does not satisfy: 31% of its parents sit on a repeated composition, and those clouds are still disconnected (Appendices A.6 and B.4). Nothing above predicts the *magnitude* of a correlation or the calibration of $\log q_\theta$ (Appendix A.8), which are separate quantities measured separately in Section 4.3.

The design criterion. The analysis therefore turns an ambiguity into a design choice. Connectivity is a property of the supervision graph, not of the model or the potential: local groups suffice for local ordering, which is what we build and measure, while controlling relative basin mass requires either a connected supervision graph — groups tiled along a path between basins, so part (ii) propagates one constant across them — or an additional global objective. This paper supplies the first and names the second.

4 EXPERIMENTS

Evidence comes from two tracks. **Track A**, a matched mechanism grid, asks what inside the value term carries the gain; **Track B**, a conservative held-out endpoint, asks whether that gain survives a corrected evaluation and extends to retrieval. Track A is the cleaner design for causal comparison, its cells differing only in the physics block and the seed; the family behind Track B has known configuration provenance (Appendix B.10) and is not presented as a matched trial.

Shared setting. Every arm trains the bond-free FlowMol3 backbone (Dunn & Koes, 2025) on OMol25 (Levine et al., 2025) against the OMol25/eSEN teacher (Fu et al., 2025) under the settings of Section 2, for 30 000 optimiser steps at effective batch 16, approximately 0.12 data epochs. The read-out protocol is fixed and operational: $\log q_\theta$ comes from an ODE integration of the clamped positional flow with Hutchinson divergence estimation (Grathwohl et al., 2019; Hutchinson, 1989), and every evaluation energy is a GFN2-xTB single point (Bannwarth et al., 2019) from an evaluator that enters no training loss and shares no parameters or functional form with the teacher. Each held-out parent contributes one group of $K = 9$ geometries, its data geometry plus 8 isotropic displacements at $\sigma = 0.15$ Å per coordinate: a local stress test rather than a thermal conformer ensemble, measuring no mode occupancy.

Track A: the matched mechanism grid. Six cells at five seeds each, warm-started from a common checkpoint and matched in *every* non-physics setting: optimiser, budget, batch, estimator, and shards drawn from the training split alone. The cells are **A**, flow matching only; **B**, the same penalty on shards whose energies are *identically zero*, so (2) degenerates to $\text{Var}_k[\log q_\theta]$ and no energy label enters the loss; **C**, energies scrambled within each parent, a y -randomisation (Rücker et al., 2007) carrying the treatment’s own energy multiset and spread; **D**, true energies; **E**, the gradient baseline of Section 2.5 alone; **F**, both, all scored on 120 held-out parents with the data geometry dropped.

Track B: the conservative held-out endpoint. Two features of the earlier evaluation population inflate the contrast, and this endpoint removes both: the data geometry is systematically its group’s lowest-energy member and hence high-leverage, and one perturbation shard read by every value-supervised arm in that family was built from the split the evaluation parents come from while the control reads none (Appendix B.4). Every geometry was therefore re-scored from one source, and each per-group statistic is computed on the **8 displaced geometries alone over the 93 parents outside the shard pool**; both corrections apply uniformly and both *reduce* the contrast. Five flow-matching seeds and four completed value seeds enter.

Metrics. Every statistic is computed inside a parent group and aggregated afterwards, for the reason the objective is grouped: the per-molecule constant $-\log Z(c)$ would otherwise dominate a pooled statistic (Appendix B.5). For parent m ,

$$r_m = \text{corr}_k\left(\log q_\theta(\mathbf{x}_{m,k}), -\frac{E_{m,k}}{kT}\right), \quad \text{NRV}_m = \frac{\text{Var}_k[\log q_\theta(\mathbf{x}_{m,k}) + E_{m,k}/kT]}{\text{Var}_k[E_{m,k}/kT]}. \quad (4)$$

Pearson r_m is invariant under positive affine maps of either variable (Remark 13), so it scores the *shape* of $\log q_\theta$ against energy and is blind to calibration. NRV_m is invariant to the per-group constant Proposition 1 leaves free and is anchored at both ends: a within-group Boltzmann density scores 0, a *constant* log-density exactly 1. The identity $\text{NRV}_{\min} = 1 - r_m^2$ is the best any single rescaling could reach, and kT_{eff} is the implied temperature, ideally 1.0 eV (Appendix B.7). *Retrieval* asks how often ranking a parent’s geometries by $\log q_\theta$ puts the lowest-energy one first, or in the top three.

Reporting. We report effect sizes, means and standard errors over independently trained seeds, together with every seed-level result. Because several cells contain only two to five completed seeds and completion is objective-dependent, we use these descriptive quantities rather than treating asymptotic null-hypothesis tests as the primary evidence. A training run is the unit of replication, every figure draws every seed point, every value-cell mean carries its launched / completed / diverged counts, and per-seed records are in Appendices B.5, B.9 and B.15.

Two quantities run through the results below: a correlation, which scores the *direction* of the density’s response to energy, and a normalised residual variance, which also scores its *magnitude*. Value supervision moves the first and not the second. Relative basin mass, the third property of Section 3, is measured nowhere.

4.1 CORRECT ENERGY PAIRINGS DRIVE THE GAIN

Figure 2 answers three questions with one matched design. *Does the value term improve ordering?* Cell D reaches 0.421 ± 0.016 against cell A’s 0.217 ± 0.008 , $\Delta r = +0.204$, and every completed D seed lies above the observed range of every control cell. *Is the gain merely density regularisation?* Cell B, whose shard energies are identically zero so no energy label enters the loss, reaches 0.187 ± 0.030 , 0.030 below A, and D exceeds it by +0.234. *Does it require the correct correspondence?* Cell C, carrying real energies whose pairing to geometries is destroyed, sits at 0.185 ± 0.020 , again below no physics, and D exceeds it by +0.236. The three controls lie within 0.032 of one another, an order of magnitude below the treatment effect: the geometry-to-energy correspondence moves ordering, an auxiliary penalty on the log-density or the energy multiset alone does not.

The gradient comparator. Cell E carries the score-matched force term alone, completed 5/5 at 0.085 ± 0.016 , below the physics-free baseline by -0.132 ; added to the value term it subtracts again, F scoring 0.194 ± 0.016 against D’s 0.421, a difference of -0.227 . Cell E2, identical to E at $\lambda_1 = 0.01$, completed 5/5 at 0.080 ± 0.013 , so the result does not rest on one badly chosen weight (Appendix B.9). The claim stays narrow. Under this implementation’s endpoint parameterisation these cells match a score equal to $(1 - t)$ times (3) (Appendix B.6); what is measured is that comparator — naive, off-policy, endpoint — on this non-equilibrium corpus, and not (3) at the scale it defines. It is not a claim that force supervision is unhelpful for energy-grounded generation, which can be made to work (Guo et al., 2025; Hu, 2026); Propositions 2 and 3 name candidate mechanisms. Two forward-only diagnostics on the shared warm-start checkpoint locate the difficulty: the endpoint force is a poor target at the interpolant, where the force actually acting is nearly orthogonal to it (per-atom cosine 0.16 at $t^* = 0.85$) because the interpolant sits off the data manifold, and the force term’s parameter gradient points systematically against the flow-matching gradient while the value term’s is close to orthogonal to it. We read these as a measured mechanism for the observed difference rather than a proof (Appendix B.6).

Completion by cell. The launch record is A 5/5/0, B 5/4/1, C 5/5/0, D 5/3/2, E 5/5/0, F 5/2/3 launched / completed / diverged: none of the ten runs without the value term failed, against six of the twenty carrying it, cell B included, which implicates the reverse-time log-density objective and its gradient rather than the teacher’s energies; D and F are means over survivors. Two records bound that risk. A later, byte-identically configured block of four cell-D seeds completed in full at 0.353, 0.374, 0.452 and 0.471, and across all seven scored D runs the mean is 0.416 ± 0.017 ; that block is reported separately and never merged into the grid means. The F campaign launched 10, completed 3 and diverged 7, its three completed runs agreeing at 0.197 ± 0.009 (Appendices B.9 and B.15).

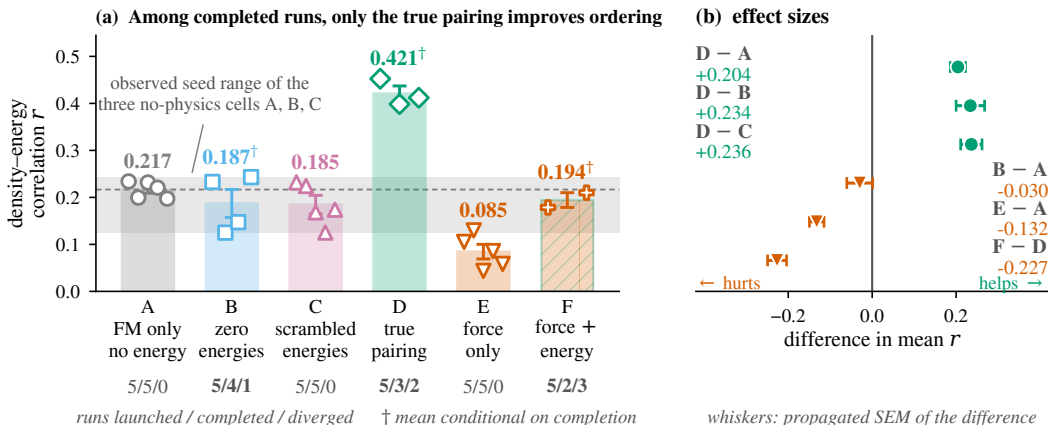


Figure 2: **A matched six-cell grid isolates what the energy term contributes.** Population: 120 held-out parents, reference geometry dropped, 8 displaced geometries each, scored by an independent GFN2-xTB evaluator, not the eSEN training teacher. Metric: the per-parent density-energy correlation r , per cell as the mean over completed training seeds (whisker: standard error over seeds) with every completed seed drawn, and in (b) as effect sizes: differences of cell means, whiskers the propagated descriptive SEM of the difference rather than a confidence interval. Cells A–F are defined in Section 4.1; runs launched / completed / diverged are printed under each cell, because completion is objective-dependent. Only the true geometry-energy pairing separates: every completed D seed lies above the shaded band, the observed seed range of A, B and C; the zero-energy and scrambled cells do not, and the gradient baseline falls below. **Cells B, D and F lost seeds to divergence** (†), so their means are conditional on completion and the D and F unconditional magnitudes are not identified. Panel detail in Appendix B.18.

4.2 THE GAIN GENERALISES TO HELD-OUT MOLECULES AND RETRIEVAL

Ordering improves well beyond seed spread. On the conservative held-out endpoint — 93 parents disjoint from the supervision pool, data geometry dropped, every geometry scored by the independent evaluator — the value term raises the mean per-parent correlation between $\log q_\theta$ and $-E/kT$ from 0.200 ± 0.018 over five flow-matching seeds to 0.369 ± 0.023 over four completed value seeds, an effect size of $\Delta r = +0.169$ (Figure 3a). All four value seeds exceed all five control seeds, so the observed seed ranges of the two arms do not overlap. The launch record is 5/5/0 for flow matching and 5/4/1 for the value arm, so the value mean is conditional on four of five completed launches, and Appendix B.5 reports conservative imputations for that run. The shift is spread across the population rather than carried by a few parents, value supervision scoring higher on 67 of the 93 (Figure 3b).

Retrieval improves, and the gain persists at fixed RMSD. Ranking a parent’s eight displaced geometries by $\log q_\theta$ identifies the lowest-energy one 29.6% of the time under value supervision against 20.2% for the control, with 12.5% at chance, and top-3 rises from 49.5% to 62.4% against 37.5% (Figure 3c). Retrieval is invariant to the per-group constant and to calibration. The gain also persists at fixed RMSD: on a family holding the within-group displacement magnitude constant to numerical precision, value supervision reaches 0.305 against 0.182, a gap of +0.124 with four seeds per arm (Figure 3d), which rules out radial-distance variation as the sole explanation. A normal-mode family also has a positive mean gap (+0.185 over five seeds per arm), but its observed seed ranges overlap, so we treat it as supporting context only (Appendix B.8).

Where the level sits. Teacher and evaluator are different theories of chemistry, so a model reproducing its teacher exactly would not score 1: pushing the *teacher’s own* energies through the identical pipeline gives $r = 0.927$ on the same 93 parents, an ensemble-specific reference, not a ceiling (Appendix B.7). Budget and locality remain open explanations of the rest.

4.3 VALUE SUPERVISION IMPROVES ORDERING BEFORE CALIBRATION

Direction improves; the raw scale does not. Ordering and calibration are different requests of the same density, and the objective clearly answers the first. On the same models, seeds and parents, the normalised residual variance rises from 1.402 to 2.144, where a *constant* log-density scores exactly 1

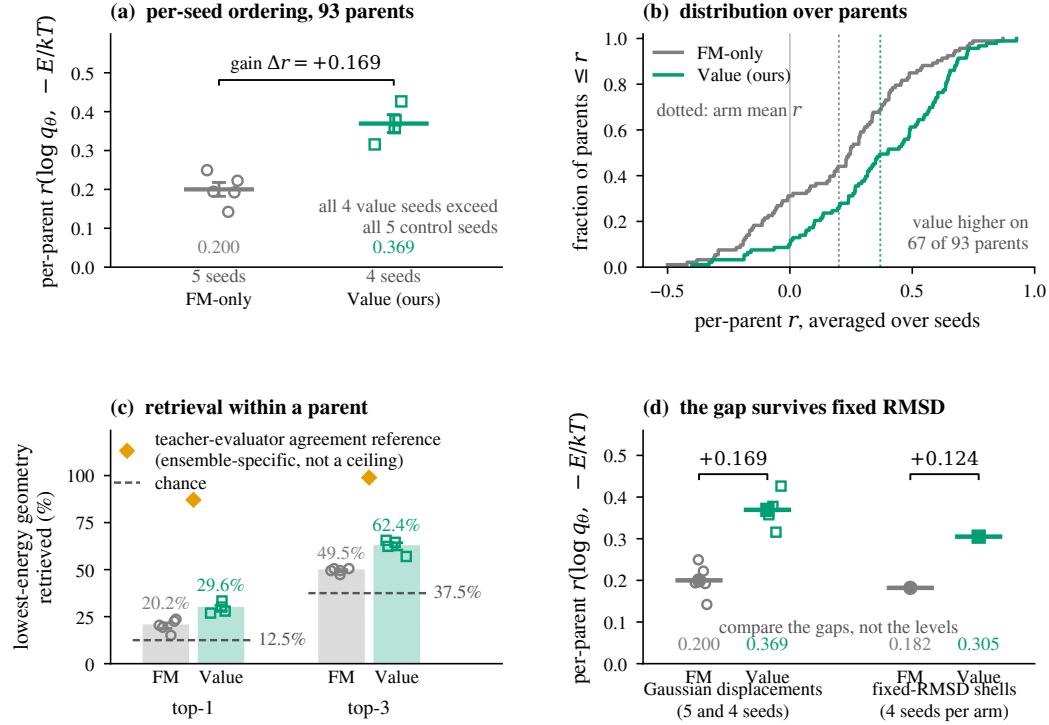


Figure 3: **Energy-value supervision improves local density-energy ordering.** Population for (a)–(c): the 93 held-out parents disjoint from the value term’s shard pool, data geometry dropped, up to eight displaced geometries each, scored by GFN2-xTB, an evaluator independent of the eSEN potential that supplied the training energies. The metric is the per-parent correlation between the clamped positional-flow density $\log q_\theta$ and $-E/kT$. (a) Every seed of both arms; rules are arm means, bars their standard error over seeds, and the gain is $\Delta r = +0.169$ with all four value seeds above all five control seeds. The value arm is 5/4/1 launched / completed / diverged, so its mean is conditional on completion. (b) The shift is spread across the population rather than carried by a few parents. (c) Ranking a parent’s geometries by $\log q_\theta$ retrieves the lowest-energy one more often; the orange marker is the teacher-evaluator agreement reference (not a ceiling): that evaluator scoring the training teacher, ensemble-specific and re-measured for any other ensemble. (d) Holding the within-group displacement magnitude constant to numerical precision leaves the gap in place, so distance variation is not sufficient to explain the gain; that family is a separate evaluation round on its own parents with four seeds per arm, and only its arm means are plotted, so the comparison to make is between the two gaps, not between the two levels. Panel detail in Appendix B.18.

(Figure 4a): the arm that ranks geometries better sits further from the Boltzmann scale, which is why we report the two quantities separately rather than as one claim of physical grounding.

The usable signal nonetheless increases. No rescaling can bring a parent’s NRV_m below $1 - r_m^2$, and that optimally rescaled bound falls from 0.825 to 0.747 (Figure 4b), so the density carries more energy-aligned information than before. The implied temperature moves the same way, from $kT_{\text{eff}} = 8.02$ eV to 2.34 eV against the 1.0 eV target (Figure 4c), and so does the slope, from 0.130 to 0.448: a response that was far too flat moves to the right order of magnitude, overshooting rather than approaching. The same separation appears against the eSEN training teacher, where r rises from 0.202 to 0.399 while raw NRV rises from 1.676 to 2.821 (Appendix B.7).

What a further intervention has to control. Raw NRV additionally charges for the part of the residual that is not energy-aligned, and that part grows enough to push NRV up even as the rescaled bound falls; the free constant per group is not the cause, since NRV is invariant to it. Repair requires an additional scale-sensitive objective controlling the magnitude and the energy-unaligned component of the response — a different term from (2), and one this paper does not run. On generation itself we detect *no degradation in validity*, *PoseBusters pass rate* (Buttenschoen et al., 2024) or *uniqueness* at the evaluated sample size (Appendix B.13).

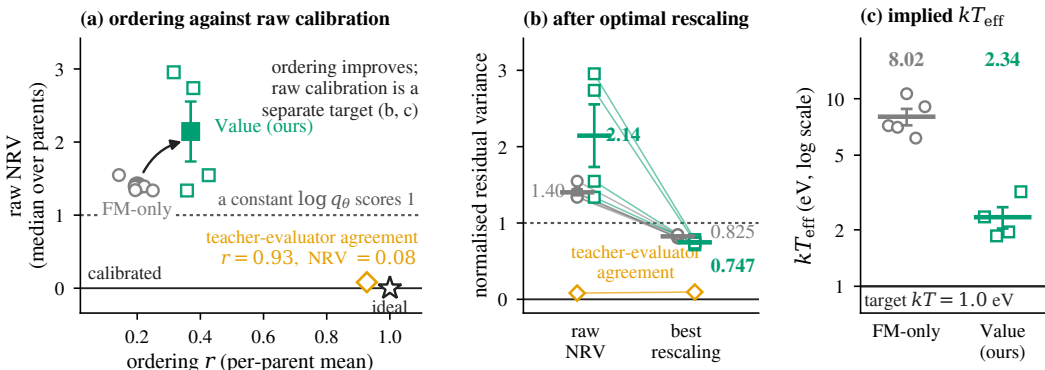


Figure 4: **Value supervision improves ordering before calibration.** Same 93 held-out parents as Figure 3, reference geometry dropped, GFN2-xTB evaluator; five seeds for the flow-matching control, four for the value arm (open markers; filled marker or rule the arm mean, bar its standard error over seeds). Metrics: ordering r against raw calibration $\text{NRV} = \text{Var}(\log q_\theta + E/kT)/\text{Var}(E/kT)$ (a), the optimally rescaled residual $\text{NRV}_{\min}$ (b), and the implied kT_{eff} (c). Three diagnostics move toward the Boltzmann-consistent regime: r rises from 0.200 to 0.369, $\text{NRV}_{\min}$ falls from 0.825 to 0.747, and kT_{eff} moves from 8.02 to 2.34 eV toward the $kT = 1.0$ eV target. Raw residual variance — the diagnostic most directly aligned with the objective as displayed — remains a separate unmet target: NRV rises from 1.402 to 2.144, past the 1 that a constant log-density scores, showing that errors in response magnitude and energy-unaligned variation remain even as ordering improves. The value arm (5/4/1 launched / completed / diverged) is conditional on completion. The orange marker is the teacher-evaluator agreement reference (not a ceiling); panel detail and its derivation are in Appendix B.18.

5 CONCLUSION

Energy values from a transferable universal potential are usable supervision for a de novo molecular generator. One group-wise variance term, costing pre-computed labels and no partition function, raises held-out local density–energy ordering from 0.200 to 0.369 and top-1 retrieval from 20.2% to 29.6% under an independent scorer, and a matched grid places that gain in the correct geometry–energy correspondence rather than in an auxiliary penalty.

The same analysis bounds what it buys. What we measure is a finite-step estimate of the clamped positional-flow density q_θ , which need not equal the generator’s joint conditional $p_\theta(x | c)$ (Assumption 2); the objective is also numerically demanding, so every value-cell mean is conditional on completion and carries its launched, completed and diverged counts (Appendix B.15). And it is local: inside a group the density–energy relation is fixed only up to one constant per connected component and our groups are disconnected (Proposition 1), so relative basin mass is not identified and raw calibration is not achieved (Appendix B.17 has the full inventory). Local ordering, calibration and global allocation are three targets rather than one, and we deliver the first with a design criterion for the rest — connect the groups, or add a global term.

STATEMENT ON THE USE OF LARGE LANGUAGE MODELS

Generative AI tools were used throughout the preparation of this work. They assisted with software implementation and debugging, plotting, construction of the result inventory, literature organisation, manuscript structuring and language editing, and they provided feedback on experimental design and interpretation and help in drafting the mathematical exposition. The authors selected the research questions and the final methodology, executed or supervised every experiment, inspected the per-run records, verified every reported number against its source file, checked the mathematical statements, reviewed the cited literature, and edited all AI-assisted text. No result or claim in this paper is included on the authority of an AI system alone. The authors take responsibility for the complete manuscript and for the accompanying artifacts.

REPRODUCIBILITY STATEMENT

What the paper specifies. The objective is stated in full in Section 2 ((1), (2)), and the gradient actually taken through it — a truncated-adjoint surrogate rather than differentiation through the

solve — is written out in Appendix B.1. The standing assumptions and the complete proofs behind Proposition 1, in particular Theorem 2 and Corollary 2, are in Appendix A. The backbone, optimiser, schedule, objective weights, warm-up and ramp, clipping thresholds, probe times and estimator settings are tabulated in Appendix B.2; the training scale and hardware are in Appendix B.3; data preprocessing and the construction of the perturbation shards, including which shards each arm reads, are in Appendix B.4. The evaluation protocol, the independent scorer, the primary-endpoint construction and the aggregation rules are in Appendices B.5 and B.7, which also state the reporting policy — effect sizes, arm means with standard errors over independently trained seeds, and every seed-level result — and which give the adverse imputation for the one unreported run. The inferential statistics computed during the analysis are collected once, for completeness and in one place, in Appendix B.16. The complete launch record, including every run that diverged, every run that completed without being scored, and the later seeds launched outside the matched design, is in Appendices B.9, B.10 and B.15, and Appendix B.12 inventories every evaluation we ran, scored or not. Appendix B.18 states how each main-text figure panel was constructed from those records.

What the displayed commands do and do not reproduce. Appendix B.19 gives the two environments, why they are separate, and the ordered command sequence. That sequence is exact for one of the paper’s two families of numbers and not for the other, and we state the difference plainly. The matched six-cell grid of Section 4.1 — and hence Figure 2 and the per-seed tables of Appendix B.9 — follows from the commands as displayed: perturbation shards drawn from the training split alone, $M = 8$ parents per value step, value-loss cap 1500. The primary held-out endpoint of Sections 4.2 and 4.3 does *not*: those checkpoints come from an earlier family trained under the archived configurations described in Appendix B.10, whose shard list also contains a validation-split shard, with $M = 4$ parents per value step and a value-loss cap of 3000, evaluated on 120 parents and then restricted by the disjoint-parent filter to the 93 reported. Reproducing that endpoint therefore means training from those archived configurations rather than from the matched-grid ones; running the displayed training commands and applying the endpoint analysis would reproduce the protocol but not the checkpoints. Appendix B.10 records that family’s configuration defects and states which of its contrasts survive the corrections and which are superseded by the matched grid.

Two caveats a re-runner should expect. Diverged runs are recorded rather than re-rolled, and because divergence is objective-dependent, every value-cell mean is reported conditional on completion with its launched, completed and diverged counts beside it (Appendix B.15); a re-run at a different seed set should expect a different completion pattern. And every reported density is a finite-step estimate: $\log q_\theta$ comes from an ODE integration of the clamped positional flow with Hutchinson divergence estimation, whose settings are tabulated in Appendix B.2.

ETHICS STATEMENT

This work trains a generative model of small molecules against a publicly released quantum-chemistry corpus and a publicly released neural potential, and evaluates it with an open-source semi-empirical method; no human subjects, personal data or human annotation are involved. De novo molecular generators carry a dual-use risk in common with the rest of this literature, in that a method for proposing molecules with desired properties could in principle be directed at harmful ones. The models reported here are trained for approximately 0.12 epochs, are not conditioned on any biological or toxicological objective, and generate chemically valid structures at a rate well below what a practical design pipeline would require, so we judge the incremental risk of this particular study to be low. The scientific claims are deliberately bounded: we report a local ordering result, report calibration and global mass allocation as separate and unmet targets, and Section 5 states what the paper does not establish.

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A THEORY: ASSUMPTIONS, ESTIMATOR, AND PROOFS

This appendix collects everything the main text states without proof, in the order the argument uses it: the standing assumptions (Appendix A.1); what the reported log-density actually is, derived from the flow (Appendix A.2); the score read-out the gradient baseline relies on and the two reasons it is a poor target (Appendices A.3 and A.4); the certificate behind the value objective and exactly what it identifies (Appendices A.5 and A.6); the identity linking the training objective to the reported metric (Appendix A.7); and a closing inventory of which statements this paper exercises and which it only states (Appendix A.8).

A.1 SETUP, NOTATION AND STANDING ASSUMPTIONS

A molecular state is a pair $(c, \mathbf{x})$. Operationally the composition $c \in \mathcal{C}$ is the element multiset with the atom count $N(c)$ it determines and the total charge q , which is what the pipeline threads to the teacher, and $\mathcal{C}$ is countable. Spin multiplicity s is a standing restriction rather than a component of c : it is never passed and the teacher defaults to a singlet (Assumption 6), so every statement below is at that default and the composition audit used for identifiability keys on (element multiset, charge) accordingly. The conformation $\mathbf{x} \in \mathbb{R}^{3N(c)}$ is the nuclear coordinate vector with the centre of mass removed, so the effective dimension is $d = 3(N - 1)$. The teacher is a universal neural potential $E : (c, \mathbf{x}) \mapsto \mathbb{R}$ (Levine et al., 2025; Fu et al., 2025), smooth in $\mathbf{x}$, with conservative forces $F(c, \mathbf{x}) = -\nabla_{\mathbf{x}} E(c, \mathbf{x})$, and $kT > 0$ is a fixed temperature scale in eV. Throughout, F denotes the force and never a free energy.

The target is the joint Boltzmann law and its two factors,

$$\pi(c, \mathbf{x}) = \frac{e^{-E(c, \mathbf{x})/kT}}{\mathcal{Z}}, \quad Z(c) = \int e^{-E(c, \mathbf{x})/kT} d\mathbf{x}, \quad A(c) = -kT \log Z(c), \quad (5)$$

so that $\mathcal{Z} = \sum_c Z(c) = \sum_c e^{-A(c)/kT}$, $\pi(c) = e^{-A(c)/kT}/\mathcal{Z}$ and $\pi(\mathbf{x} | c) = e^{-E(c, \mathbf{x})/kT}/Z(c)$. Neither $Z(c)$ nor $\mathcal{Z}$ is ever formed. The generative model is a flow $d\mathbf{x}_t/dt = v_\theta(\mathbf{x}_t, t)$ transporting $p_0 = \mathcal{N}(0, \sigma^2 I)$ on the centre-of-mass-free subspace at $t = 0$ to $q_\theta := q_1^\theta$ at $t = 1$ with the discrete channels clamped at their endpoint labels (Assumption 2), trained on the linear interpolant $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$ (Lipman et al., 2023; Liu et al., 2023; Albergo et al., 2023).

Assumption 1 (Teacher regularity). $E(c, \cdot)$ is the teacher potential at fixed (N, q, s) and is C^1 in $\mathbf{x}$ on a connected domain.

Assumption 2 (Which density is reported: the clamped positional-flow density). The quantity every theorem below is stated for, and every experiment reports as $\log q_\theta(\cdot | c)$, is the log-density of the *clamped positional flow*: the flow obtained by holding the discrete channels — atom types, formal charges and the empty bond tensor — at their endpoint labels (a_1, c_1, e_1) for the whole reverse trajectory, while only $\mathbf{x}$ evolves. That object is positive on the sampled support and normalised on the centre-of-mass-free subspace by construction, being an exact change of variables applied to a normalised Gaussian prior (Grathwohl et al., 2019), up to the discretisation of Assumption 4. It coincides with the generator’s own joint conditional density $p_\theta(\cdot | c)$ only if the discrete trajectory is determined by its endpoint; the generator evolves types and charges through a continuous-time Markov chain, so in general the two differ, we have not quantified the difference, and this is recorded as a limitation (Appendix B.17). The symbol q_θ is used nowhere else in this appendix.

Assumption 3 (Translation-invariant velocity field). v_θ is invariant to global translation, so the ambient $3N$ -dimensional trace of its Jacobian equals the intrinsic $3(N - 1)$ -dimensional one (Appendix A.2). The centre-of-mass-removed equivariant field of the backbone satisfies this. **Rotations are not quotiented**: the prior log-density uses effective dimension $3(N - 1)$, i.e. centre-of-mass removal only. We do not claim a rotation-quotiented density.

Assumption 4 (Discretised likelihood). The reported $\log q_\theta$ is a discretised continuous-flow likelihood — explicit Euler in the state, midpoint-rule quadrature for the divergence integral, with a

stochastic trace estimator — so it is a biased and noisy estimate of the exact clamped positional-flow log-density. The discretisation controls that bias; it does not change which density is estimated. The scheme is derived in Appendix A.2.

Remark 1 (What the theory is scoped to, given the implemented gradient). The gradient actually back-propagated through $\mathcal{L}_{\text{value}}$ is a truncated-adjoint surrogate: the reverse trajectory is advanced without gradient tracking, each state is detached before its divergence is taken, and the prior term is computed from a detached endpoint, so the parameter gradient flows only through the divergence evaluations (Appendix B.1). We do not differentiate through the ODE solve, and the implemented update is therefore not exact likelihood-gradient descent. Every statement in this appendix is accordingly scoped to the *zero set and identifiability* of the ideal residual condition $w \equiv \text{const}$ — what a model that had driven the population objective to zero would have been shown, and what that leaves free — and to nothing about whether, or to what, the implemented update converges.

Assumption 5 (Rigid-motion and symmetry factors). $Z(c)$ in (5) is an integral over $\mathbb{R}^{3N}$ and therefore carries translational and rotational volume factors and a symmetry-number correction. We take $Z(c)$ to be the configurational integral in the centre-of-mass-removed space that the density estimator actually models; as noted in Assumption 3, rotations are *not* quotiented by the implementation. Within a parent group this is benign: all K perturbations are small displacements of one data geometry in one frame, so any rotational volume factor is approximately constant within the group and cancels from the per-group correlation, and is exactly annihilated by the within-parent variance loss. It is *not* benign for any comparison across groups or across separated basins, which is one more reason we make none (Theorem 2).

Assumption 6 (Well-posed conditional energy). $E(c, \cdot)$ is the potential at fixed (N, q, s) , so the teacher must be queried with the correct charge and spin. Our implementation decodes the total charge from the frozen charge channel and passes it to the teacher; the spin multiplicity is *not* passed and the teacher defaults to a singlet. For radicals and open-shell complexes this is the wrong electronic state. Every reported evaluation is on main-group organic parents drawn from the held-out split, where the singlet default is the standard assignment for closed-shell species; we have not verified the multiplicity of every evaluated parent. The restriction is recorded as a limitation (Appendix B.17) and is the reason no radical, open-shell or transition-metal result is reported.

A.2 THE REPORTED LOG-DENSITY: CHANGE OF VARIABLES, TRACE ESTIMATION, AND THE INTEGRATION SCHEME

Everything the value objective constrains, and everything the calibration diagnostics measure, is the number this subsection defines. It is derived in three steps — an exact identity, an unbiased estimator of the one expensive term in it, and an integration scheme — and the third step is the one that decides how *accurately* the density is reported, not which density it is.

Step 1: the instantaneous change of variables. Let $\mathbf{x}_t$ solve $d\mathbf{x}_t/dt = v_\theta(\mathbf{x}_t, t)$ and let p_t be the law of $\mathbf{x}_t$. Mass conservation is the continuity equation $\partial_t p_t + \nabla \cdot (p_t v_\theta) = 0$. Expanding the divergence, $\nabla \cdot (p_t v_\theta) = \nabla p_t \cdot v_\theta + p_t \nabla \cdot v_\theta$, and dividing by $p_t > 0$,

$$\partial_t \log p_t + v_\theta \cdot \nabla \log p_t = -\nabla \cdot v_\theta. \quad (6)$$

The left-hand side is the material derivative of $\log p_t$ along a trajectory, so

$$\frac{d}{dt} \log p_t(\mathbf{x}_t) = -\nabla \cdot v_\theta(\mathbf{x}_t, t), \quad (7)$$

the instantaneous change of variables underlying continuous normalising flows (Grathwohl et al., 2019). Integrating (7) from $t = 0$ to $t = 1$ along the trajectory that ends at $\mathbf{x}_1$ and using $p_0 = \mathcal{N}(0, \sigma^2 I)$ gives (1) of the main text,

$$\log q_\theta(\mathbf{x}_1 | c) = \log p_0(\mathbf{x}_0) - \int_0^1 \nabla \cdot v_\theta(\mathbf{x}_t, t) dt, \quad \mathbf{x}_0 = \mathbf{x}_1 - \int_0^1 v_\theta(\mathbf{x}_t, t) dt. \quad (8)$$

In practice the trajectory is generated *backwards* from the data point $\mathbf{x}_1$, so one reverse solve returns both $\mathbf{x}_0$ and the accumulated divergence. Equation (8) is exact in continuous time and requires no invertibility argument beyond uniqueness of the ODE solution; every approximation in the reported number enters in steps 2 and 3.

Step 2: the trace, and why a stochastic estimator is used. The divergence is the trace of the Jacobian, $\nabla \cdot v_\theta = \text{tr } J$ with $J_{ij} = \partial(v_\theta)_i / \partial x_j$. Computing $\text{tr } J$ exactly costs $3N$ backward passes, one per coordinate, at every integration step. The Hutchinson estimator (Hutchinson, 1989) replaces it by

$$\text{tr } J = \mathbb{E}_\epsilon [\epsilon^\top J \epsilon], \quad \mathbb{E}[\epsilon] = 0, \quad \mathbb{E}[\epsilon \epsilon^\top] = I, \quad (9)$$

which is exact in expectation for any such ϵ , and each sample costs one vector–Jacobian product — a single backward pass — followed by an inner product with ϵ . We use $n_p = 2$ Rademacher probes per step, i.e. two backward passes in place of $3N$.

Two properties of (9) matter for how the resulting numbers may be read. First, with independent entries the estimator’s variance is

$$\text{Var}[\epsilon^\top J \epsilon] = \sum_{i < j} (J_{ij} + J_{ji})^2 \xrightarrow{J=J^\top} 2 \left(\|J\|_F^2 - \sum_i J_{ii}^2 \right), \quad (10)$$

because $\epsilon^\top J \epsilon = \sum_i J_{ii} + \sum_{i < j} (J_{ij} + J_{ji}) \epsilon_i \epsilon_j$ and the products $\epsilon_i \epsilon_j$ over distinct pairs are uncorrelated with unit variance. Rademacher probes are the standard choice because the diagonal of J contributes no variance at all, whereas Gaussian probes pay $2\|J\|_F^2$; averaging n_p probes divides (10) by n_p . Second, the divergence enters (8) *linearly*, so trace noise adds variance to the reported $\log q_\theta$ but no bias: an unbiased trace at every step gives an unbiased estimate of the discretised integral. The bias in the reported number comes from the quadrature, not from the probes. We have not measured the repeat-to-repeat spread of $\log q_\theta$ on the reported population, and say so as a limitation (Appendix B.17).

Finally, Assumption 3 is what lets the probes be drawn in the ambient $3N$ coordinates without correcting the dimension. If v_θ is invariant to a global translation then $Ju = 0$ for each of the three uniform-translation directions u , so those directions contribute zero eigenvalues and the ambient trace equals the trace of J restricted to the centre-of-mass-free subspace. Rotations enjoy no such identity and are not quotiented anywhere in the implementation.

Step 3: the integration scheme, and what the discretisation does and does not control. The reverse-time solve of (8) uses *explicit Euler in the state with midpoint-rule quadrature for the divergence integral*. With n steps the implementation sets $\Delta t = 1/n$ and evaluates the field and its divergence at

$$t_k = 1 - \left(k + \frac{1}{2}\right) \Delta t, \quad k = 0, 1, \dots, n-1, \quad (11)$$

accumulating $-\Delta t \nabla \cdot v_\theta(\mathbf{x}_{t_k}, t_k)$, advancing the state by $\mathbf{x} \leftarrow \mathbf{x} - \Delta t v_\theta(\mathbf{x}, t_k)$, and adding the prior log-density at the end.

Two schemes are in play and they must not be conflated. The *state* update is explicit Euler with n steps of size $1/n$, which traverses the whole of $[0, 1]$ and carries $O(\Delta t)$ error. The *divergence* integral is approximated by the midpoint rule on $[0, 1]$: all n nodes are interior to the interval, the interval itself is not truncated, and the rule carries $O(\Delta t^2)$ error and is exact for integrands linear in t . The half-step offset therefore avoids evaluating the field at $t = 1$ without moving the endpoint the scheme targets. What the estimator targets is $\log q_\theta$ at $t = 1$; there is no family of smoothed marginals indexed by the discretisation, and a finer grid estimates the *same* quantity more accurately rather than a different one. Analytic checks on the implementation confirm this: on a field of constant divergence, and on a time-varying one, the accumulated integral equals the full-interval value at every n rather than a truncated $1 - 1/(2n)$ value, and on a linear continuous normalising flow the estimate converges to the closed-form endpoint log-density as n grows.

What the finite grid costs is accordingly *numerical discretisation error*, dominated by the $O(\Delta t)$ Euler state integration rather than by the $O(\Delta t^2)$ quadrature, together with whatever endpoint stiffness the learned field carries.

Two uses of the symbol ϵ must be kept apart, and neither is a smoothing scale of the reported density: the ϵ of (15) below is the Gaussian width $(1-t)\sigma/t$ attached to the flow-matching score read-out at a probe time t and not to the likelihood estimator, and it is unrelated to the mass-relaxation ε of Remark 9, which is written with a different glyph throughout.

A.3 THE FLOW-MATCHING SCORE READ-OUT

The gradient-supervision baseline of Section 2 needs the score of the model’s own time- t marginal. Flow matching supplies it in closed form, which is convenient and, as Appendix A.4 shows, also the source of the baseline’s trouble.

Lemma 1 (Flow-matching score read-out). *Let $x_t = (1 - t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, \sigma^2 I)$ independent of $x_1 \sim p_{\text{data}}$ and $v^*(x, t) = \mathbb{E}[x_1 - x_0 \mid x_t = x]$. Then for every $t \in [0, 1)$, $\nabla_x \log p_t(x) = (tv^*(x, t) - x)/((1 - t)\sigma^2)$.*

Proof of Lemma 1. Conditionally on the data endpoint, $x_t \mid x_1 \sim \mathcal{N}(tx_1, (1 - t)^2\sigma^2 I)$, so the time- t marginal $p_t(x) = \int \mathcal{N}(x; tx_1, (1 - t)^2\sigma^2 I) p_{\text{data}}(x_1) dx_1$ is a Gaussian mixture with covariance independent of the mixing variable. Differentiating under the integral sign and using $\nabla_x \mathcal{N}(x; \mu, \Sigma) = -\Sigma^{-1}(x - \mu)\mathcal{N}$ gives the posterior-mean form

$$\nabla_x \log p_t(x) = \frac{t \mathbb{E}[x_1 \mid x_t = x] - x}{(1 - t)^2\sigma^2}. \quad (12)$$

The flow-matching regression target is $x_1 - x_0$, and $x_0 = (x_t - tx_1)/(1 - t)$ gives $x_1 - x_0 = (x_1 - x_t)/(1 - t)$. Taking conditional expectations, the marginal velocity is $v^*(x, t) = (\mathbb{E}[x_1 \mid x_t = x] - x)/(1 - t)$, i.e. $\mathbb{E}[x_1 \mid x_t = x] = x + (1 - t)v^*(x, t)$. Substituting into (12), one factor of $(1 - t)$ cancels and

$$s^*(x_t, t) = \frac{tv^*(x_t, t) - x_t}{(1 - t)\sigma^2}. \quad (13)$$

Replacing v^* by the learned v_θ defines s_θ . Finally, the map $v \mapsto s$ in (13) is affine with linear part $t/((1 - t)\sigma^2)$, so a velocity perturbation δv moves the implied score by $t\|\delta v\|/((1 - t)\sigma^2)$. $\square$

Remark 2 (Sign). Matching $s_\theta \rightarrow +F/kT$ is Boltzmann-attracting because $\nabla_x \log \pi = -\nabla_x E/kT = +F/kT$. The sign in (13) is not a convention that may be flipped.

Remark 3 (The read-out as implemented). Lemma 1 takes a velocity. Under the parameterisation this implementation uses, the network’s position output is the endpoint prediction rather than a velocity, and the gradient-supervision term passes that output to (13), so the score this comparator matches is $(1 - t)$ times the one (13) defines. Every force number in this paper is produced with that read-out. Appendix B.6 specifies it in full and reports what the diagnostics, run under both read-outs, say about it.

Remark 4 (Amplification at the probe times used). With $\sigma = 1$, the factor $t/((1 - t)\sigma^2)$ equals 5.7, 11.5 and 32.3 at the three training probe times $t \in \{0.85, 0.92, 0.97\}$. The pole at $t = 1$ is not a removable artefact: s_θ is the score of the *smoothed* marginal p_t , and the smoothing vanishes exactly as the read-off amplification diverges.

A.4 WHY THE GRADIENT CONDITION IS A TRAPPED SURROGATE

Two mechanisms, both specific to matching (13) *off-policy at data endpoints*, are compatible with the gradient baseline failing to help. Both turn on one structural fact: the network is a function of the interpolation state x_t while the target is the force read at the data endpoint x_1 . Both are stated as *diagnostic hypotheses* rather than as explanations of the measurement in Section 4.1; neither is isolated experimentally, and neither bears on force supervision in general (Guo et al., 2025; Hu, 2026).

Proposition 2 (Bias–variance trap: a diagnostic hypothesis). *At the population optimum of $\mathcal{L}_{\text{FM}}$, and with $\epsilon = (1 - t)\sigma/t$, the read-out is, up to a factor $1/t$, the score of $p_{\text{data}} * \mathcal{N}(0, \epsilon^2 I)$ evaluated at $x_1 + \epsilon z$, $z \sim \mathcal{N}(0, I)$. Compared with the endpoint target $F(x_1)/kT$ this incurs a scale error and a smoothing error, each $O(1 - t)$ or smaller, together with an evaluation-point displacement of size ϵ , while by Remark 4 the gradient noise injected into the parameters grows as $(1 - t)^{-2}$. No choice of t makes the deterministic terms and the noise term small together.*

Proposition 3 (Objective conflict: a diagnostic hypothesis). *The population minimiser of $\mathcal{L}_{\text{FM}}$ has implied score $\nabla \log p_t^{\text{data}}(x_t)$. The network input is x_t while the force target is the endpoint quantity $F(c, x_1)/kT = \nabla \log \pi(x_1)$, so the population minimiser of the idealised squared off-policy force loss is not $\nabla \log \pi(x_1)$ but the conditional expectation*

$$s_\theta(x_t, t) \longrightarrow \mathbb{E}[\nabla \log \pi(x_1) \mid x_t, t, c] = \mathbb{E}[F(c, x_1)/kT \mid x_t, t, c], \quad (14)$$

a smoothed and shrunk field. The two population targets differ whenever the corpus is not drawn from the target law, $p_{\text{data}} \neq \pi$, and the joint minimiser is neither; a fixed λ_1 moreover weights the conflicting target by $(1-t)^{-2}$.

Proof of Proposition 2 and Proposition 3. Reparameterise the smoothing. Write $\mathbf{x}_t = t\mathbf{y}$ with

$$\mathbf{y} = \mathbf{x}_1 + \epsilon \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(0, I), \quad \epsilon = (1-t)\sigma/t, \quad (15)$$

so that $p_t(\mathbf{x}) = t^{-d} \tilde{p}_\epsilon(\mathbf{x}/t)$ with $\tilde{p}_\epsilon = p_{\text{data}} * \mathcal{N}(0, \epsilon^2 I)$, and therefore $s_\theta(\mathbf{x}_t, t) = t^{-1} \nabla \log \tilde{p}_\epsilon(\mathbf{x}_1 + \epsilon \mathbf{z})$ at the population optimum. Against the target $\nabla \log \pi(\mathbf{x}_1)$ this exposes four mismatches: a $1/t$ prefactor contributing relative error $(1-t)/t = O(1-t)$; a smoothing error $\nabla \log(p * \mathcal{N}_\epsilon) - \nabla \log p = O(\epsilon^2) = O((1-t)^2)$ by the heat-flow expansion; an evaluation-point displacement contributing $\epsilon \nabla^2 \log \pi \mathbf{z}$ to first order; and, since by Lemma 1 any velocity error is multiplied by $t/((1-t)\sigma^2)$ and the loss is quadratic in the score, parameter-gradient noise scaling as $(1-t)^{-2}$. The first three shrink as $t \rightarrow 1$ while the fourth diverges, so the total error is bounded below by a trade-off of the form $O(1-t) + O((1-t)^{-2}\eta)$ with η the velocity-estimation noise level, which is Proposition 2; averaging over probe times, as we do, reduces the constant but not the trade-off.

For Proposition 3, the displacement is not merely noise that averages away. The loss is minimised over functions of $\mathbf{x}_t$, and $(\mathbf{x}_t, \mathbf{x}_1)$ are dependent but not mutually determined, so for a squared loss the population minimiser over measurable functions of the input is the conditional expectation (14), by the orthogonality property of L^2 projection; it equals $\nabla \log \pi(\mathbf{x}_1)$ only if $\mathbf{x}_1$ is $\sigma(\mathbf{x}_t)$ -measurable, which the interpolation makes false for every $t < 1$. The result is a shrinkage *bias* in the learned score rather than a variance term, and the two population targets — the score of the data interpolation marginal, and the conditional expectation of endpoint Boltzmann scores — are distinct whenever $p_{\text{data}} \neq \pi$.

Two caveats bound both propositions. The implemented ℓ is Huberised and per-atom norm-capped (3), so its population minimiser is a robustified location functional of the same conditional law and not (14). And both arguments concern population objectives at their exact optima, so they name mechanisms the construction admits rather than ones we have isolated. $\square$

Remark 5 (The antecedent holds for our data, and on-policy variants are outside what we measure). The training corpus is a curated union of relaxation trajectories, molecular dynamics snapshots and conformer sets across many systems, charges and spins (Levine et al., 2025). It is not an equilibrium sample from a single-temperature Boltzmann distribution, and the training $kT = 1.0$ eV is not a physical temperature of the dataset, so the antecedent $p_{\text{data}} \neq \pi$ of Proposition 3 holds by construction. Querying forces at model samples rather than at data points would partially repair the evaluation-point displacement and the objective conflict, because the two targets would then refer to one distribution; it would not repair the $(1-t)^{-1}$ amplification. No such variant is measured here: the gradient-supervision cells of the matched grid (Appendix B.9) carry the off-policy endpoint term alone, whereas the earlier round’s force cells also carried an on-policy distillation term, which is why that round’s force comparisons are confounded and are superseded by cell E (Appendix B.10).

A.5 THE VALUE CONDITION: A PARTITION-FUNCTION-FREE CERTIFICATE

The certificate below is the standard zero-of-the-log-variance-divergence statement written in our conditional notation. Its structural property — that an additive constant in $\log \tilde{\pi}$, hence the partition function, cancels — is due to the authors cited in Remark 6 and is not claimed here. We reproduce the proof because Corollary 1 and Theorem 2 both refer to its internals.

Theorem 1 (Zero-variance certificate on a region). *Under the regularity assumptions of Appendix A.1 (smooth E , positive normalised q_θ on the COM-removed subspace, translation-invariant velocity field), fix a composition c , a region U and a reference measure ρ with $\text{supp } \rho = U$, and let $w(x) = \log q_\theta(x | c) + E(c, x)/kT$. Then (i) $\text{Var}_{x \sim \rho}[w] = 0$ if and only if w is ρ -a.s. constant on U ; if ρ has a positive Lebesgue density on U this says $q_\theta(\cdot | c)$ restricted to U is proportional to $e^{-E(c, \cdot)/kT}$, and if ρ is atomic on K sampled points it says $\log q_\theta$ and $-E/kT$ differ by one common constant at those K points, i.e. $K - 1$ scalar equations, with no statement about points outside the sample; (ii) if ρ has a positive Lebesgue density on U then the multiplier is fixed relative to U , $a = q_\theta(U | c)/Z_U(c)$ with $Z_U(c) = \int_U e^{-E(c, x)/kT} dx$ the integral over U and not the global $Z(c)$, so that $q_\theta(\cdot | c, U) = \pi(\cdot | c, U)$ exactly; if in addition $q_\theta(U | c) = 1$ then $a = 1/Z_U(c)$, and*

$q_\theta(\cdot | c) = \pi(\cdot | c)$ globally only when U also carries all of the target’s mass, $Z_U(c) = Z(c)$; (iii) $\text{Var}_{x \sim \rho}[w]$ is invariant to $E \mapsto E + g(c)$ and needs no evaluation of $Z(c)$ or $\mathcal{Z}$.

Corollary 1 (What the empirical loss identifies). *The empirical objective (2) has null space exactly the functions that are constant within each parent group. It therefore determines $\log q_\theta(\cdot | c)$ up to one free additive constant per group, and no further.*

Remark 6 (Attribution). $\text{Var}_\rho[w]$ with $w = \log q_\theta - \log \tilde{\pi}$ and $\tilde{\pi} = e^{-E/kT}$ is the log-variance divergence (Richter et al., 2020; Nüsken & Richter, 2021; Richter & Berner, 2024). Part (iii) — invariance to an additive constant in $\log \tilde{\pi}$, hence no partition function — is the defining structural property of that divergence and is due to those authors, not to this paper. Estimating it under a reference measure that is not the model’s own (off-policy) is likewise standard (Nüsken & Richter, 2021; Sendera et al., 2024).

Remark 7 (The certificate is about the exact clamped density; what is formed is its discretisation). Theorem 1 is stated for the exact normalised q_θ , whereas what the objective and the metric both form is its finite-step Euler/midpoint estimate (Appendix A.2). The residual the objective can enforce therefore carries discretisation bias; the target quantity is unchanged by the discretisation, and the accuracy is not.

Proof of Theorem 1. Fix c , a region U and ρ with $\text{supp } \rho = U$; let $w(x) = \log q_\theta(x | c) + E(c, x)/kT$.

(i) A random variable has zero variance if and only if it is a.s. constant, so $\text{Var}_{x \sim \rho}[w] = 0$ iff $w \equiv \kappa$ ρ -a.s. on U , and then $q_\theta(x | c) = a e^{-E(c, x)/kT}$ with the positive multiplier $a = e^\kappa$, for ρ -almost every $x \in U$. Note the sign: the residual constant enters the density through $e^{+\kappa}$, and the same holds component by component in Theorem 2. The strength of that conclusion depends on ρ . If ρ has a positive Lebesgue density on U , the identity holds Lebesgue-almost everywhere on U and q_θ restricted to U is proportional to the Gibbs weight. If ρ is the empirical measure on K sampled points, the identity holds at those K points and nowhere else is constrained: the certificate is $K - 1$ scalar equations, not a density statement, and asserting proportionality on a region from finitely many pointwise equations would be a strictly stronger claim than the variance supplies. Conversely if $q_\theta(\cdot | c) \propto e^{-E(c, \cdot)/kT}$ ρ -a.e. on U then w is constant there. Note that (i) alone is a statement *about* U : nothing outside U is constrained.

(ii) Let ρ have a positive Lebesgue density on U , so that (i) gives the identity Lebesgue-a.e. on U with a positive multiplier $a = e^\kappa$. Since $q_\theta(\cdot | c)$ is a normalised density (Assumption 2), integrating $a e^{-E/kT}$ over U gives $q_\theta(U | c) = a Z_U(c)$, where $Z_U(c) = \int_U e^{-E(c, x)/kT} dx$ is the integral over U and *not* the global $Z(c)$. Hence $a = q_\theta(U | c)/Z_U(c)$, and dividing, the conditional law is pinned exactly: $q_\theta(x | c, U) = e^{-E(c, x)/kT}/Z_U(c) = \pi(x | c, U)$. Suppose in addition $q_\theta(U | c) = 1$, so that $a = 1/Z_U(c)$: this determines the multiplier but identifies $q_\theta(\cdot | c)$ with $\pi(\cdot | c)$ globally only when the target also puts all of its mass on U , that is when $Z_U(c) = Z(c)$. Otherwise the model is the Gibbs law truncated to U and renormalised, differing from $\pi(\cdot | c)$ by the factor $Z(c)/Z_U(c)$ on U and by all of $\pi(U^c | c)$ outside it. It is exactly the proviso $q_\theta(U | c) = 1$ that our per-parent perturbation groups violate, which is what Theorem 2 quantifies, and even where it holds the step to a global statement needs the target’s mass to sit on U as well. For an atomic ρ no hypothesis of this form is available at all: a finite point set is Lebesgue-null, so $q_\theta(U | c) = 0$ for any continuous model density and part (ii) is inapplicable rather than merely unattainable. The ε -relaxation of Remark 9 repairs mass leaking outside a bounded region but not a Lebesgue-null reference measure, and we make no part-(ii) claim in the empirical reading at any ε .

(iii) $\text{Var}_{x \sim \rho}[w]$ is unchanged by $E \mapsto E + g(c)$ for any function g of composition alone, because such a shift adds a constant to w within the group; and it involves neither $Z(c)$ nor $\mathcal{Z}$, because the within-group centring subtracts whatever constant those would contribute (Remark 6). $\square$

Proof of Corollary 1. The empirical objective is $\mathcal{L}_{\text{value}} = M^{-1} \sum_m \text{Var}_k(w_{m,k})$ with $w_{m,k} = \log q_\theta(x_{m,k}) + E_{m,k}/kT$, where m indexes parent groups and k indexes the K perturbations of parent m . A within-group variance is annihilated exactly by functions that are constant within each group and by nothing else, so the null space of $\mathcal{L}_{\text{value}}$ is the set of per-group additive constants — one scalar b_m per group, not one per composition. The objective therefore determines the *shape* of $\log q_\theta$ inside each group’s support and no more. Fixing those constants requires absolute energies, which is

the role of an anchor term; the anchor was inactive in every run reported in this paper (Appendix B.1). One granularity caveat belongs with that statement: the anchor as implemented predicts its constant from element counts, atom count and total charge, so it is keyed on the *composition* and can fix at most one constant per composition, not the one constant per group this corollary identifies. Those coincide only when a composition supplies a single group, which our training pool does not satisfy for 31% of its parents (Appendix B.4). $\square$

A.6 LOCAL-TO-GLOBAL IDENTIFIABILITY

Theorem 2 characterises the null space of the group-wise objective under stated support and normalisation assumptions; connectivity of the overlap graph is what removes the per-component offsets from that null space. It does not decide whether a trained model is Boltzmann, and no reported run has a connected overlap graph. Because the statement is sensitive to the choice of reference measure, we name ours first and then separate the continuous and the empirical readings.

Definition 1 (Overlap graph). Fix a composition c with groups $j = 1, \dots, J$ and reference measures ρ_j with supports U_j , all dominated by a common σ -finite measure μ , and write $f_j = d\rho_j/d\mu$. The overlap graph $\mathcal{G}_c$ has vertices $\{1, \dots, J\}$ and an edge $\{j, j'\}$ if and only if

$$\int \min(f_j, f_{j'}) d\mu > 0, \quad (16)$$

equivalently if and only if ρ_j and $\rho_{j'}$ are not mutually singular. The criterion does not depend on the choice of μ . An edge supplies the measure $d\nu = \min(f_j, f_{j'}) d\mu$, which is nonzero and absolutely continuous with respect to both ρ_j and $\rho_{j'}$; that is what forces the two constancy statements to agree, and it is what a set merely charged by both measures does not supply. Overlap here is a property of the measures and not of the topological supports: $U_j \cap U_{j'} \neq \emptyset$ is necessary for an edge but far from sufficient.

Theorem 2 (Local-to-global identifiability). *Let the population objective $\sum_j \pi_j \text{Var}_{\rho_j}[w]$ vanish, so $w \equiv b_j \rho_j$ -a.s. on U_j for each j , and let $U_c = \bigcup_j U_j$. Then (i) $b_j = b_{j'}$ for every edge of $\mathcal{G}_c$, so the constants agree within each connected component. Suppose in addition that each ρ_j has a positive Lebesgue density on U_j . Then (ii) if $\mathcal{G}_c$ is connected, w is a single constant on U_c and the model law conditioned on U_c is the Gibbs law conditioned on U_c , $q_\theta(x | c, U_c) = e^{-E(c,x)/kT} / Z_{U_c}(c)$ with $Z_{U_c}(c) = \int_{U_c} e^{-E(c,x)/kT} dx$; the global statement $q_\theta(\cdot | c) = \pi(\cdot | c)$ follows only in the further case that U_c carries all of the mass of both laws, $q_\theta(U_c | c) = 1$ and $Z_{U_c}(c) = Z(c)$; (iii) if $\mathcal{G}_c$ has $C \geq 2$ components with regions $U^{(1)}, \dots, U^{(C)}$ and $q_\theta(U_c | c) = 1$, the zero-loss densities supported on U_c form the $(C - 1)$ -parameter family carried by positive multipliers $a_i > 0$,*

$$q_\theta(x | c) = a_i e^{-E(c,x)/kT} \quad (x \in U^{(i)}), \quad \sum_{i=1}^C a_i Z^{(i)} = 1, \quad Z^{(i)} = \int_{U^{(i)}} e^{-E(c,x)/kT} dx, \quad (17)$$

in which every vector of occupancies $(q_\theta(U^{(1)} | c), \dots, q_\theta(U^{(C)} | c)) = (a_1 Z^{(1)}, \dots, a_C Z^{(C)})$ in the open simplex is attained: the relative mass of separated basins is not identified.

Part (i) needs no density assumption and is proved below in two lines of measure theory. Parts (ii) and (iii) are the continuous corollary, where a density and a normalisation hypothesis are available; the empirical corollary (Corollary 3) is the one our runs are in, and there parts (ii) and (iii) have no counterpart, so what remains is an identifiability statement. Connectivity is sufficient and, in that sense, necessary: a disconnected graph can still admit a particular zero-loss solution whose constants happen to coincide, but only connectivity forces a single constant for every zero-loss solution.

Corollary 2 (What our design identifies). *In every run reported here each parent contributes exactly one group, and ρ_j is the empirical measure on that group’s sampled geometries: one data geometry together with its $K - 1$ displacements. Distinct parents draw their geometries independently and continuously, so with probability one no two groups share a sampled point, no edge is formed, and $\mathcal{G}_c$ is a set of isolated vertices whatever J is. Only part (i) applies. The objective constrains the local relation between $\log q_\theta$ and energy inside each cloud, is blind to that cloud’s additive offset, and constrains the density outside it, in particular the relative probability of distinct basins, not at all.*

The corollary rests on mutual singularity of the group measures and not on a one-group-per-composition design, which our training pool does not have: 31% of its parents sit on a composition shared with another parent (Appendix B.4), and those repeated clouds are still disconnected, which

sharpens the point rather than weakening it. Connectivity is therefore a property of the supervision design; groups tiled along a path between basins would make $\mathcal{G}_c$ connected, which is an experiment this paper does not contain. One further gap belongs here: the objective and the evaluation metric average over *different* clouds — different potentials, parents and displacement scales (Section 4) — so the corollary bounds what the objective can constrain and does not identify the two.

Remark 8 (Which reference measure is the operative one). What the objective (2) and the evaluation metric (4) both average over is the *empirical* measure on the sampled geometries of a parent, so the empirical corollary below is the operative one throughout. Hence U_j is a finite point set, the certificate of Theorem 1 is $K - 1$ scalar equations rather than constancy on a region, and distinct parents have almost surely mutually singular reference measures. Read instead as the full-support Gaussian those points are drawn from, no two groups of a composition would be mutually singular, every pair would therefore carry an edge, and part (ii) of Theorem 2 would apply — but no run here approaches that population objective, and reading our experiments through it would credit us with a certificate we do not have. The empirical reading is the conservative one.

Corollary 3 (Empirical corollary: what zero loss certifies at sampled points). *Let each ρ_j be the empirical measure on the K_j sampled geometries of group j , and let the objective vanish. Then (a) for each j , w takes one common value b_j at the K_j sampled points of group j , which is $K_j - 1$ scalar equations and constrains no unsampled point; (b) taking μ to be counting measure on the union of the sampled points, f_j is group j ’s atomic weight function and $\int \min(f_j, f_{j'}) d\mu > 0$ holds exactly when the two groups share at least one sampled geometry; for independently drawn continuous displacements that has probability zero, so distinct groups are almost surely mutually singular and $\mathcal{G}_c$ is edgeless; and (c) consequently the objective places no constraint relating the offsets b_j of distinct groups, and the relative mass those groups receive is unconstrained by the loss. Statement (c) is an identifiability statement and not a normalisation decomposition: the family (17) is derived only in the continuous reading, where the $U^{(i)}$ carry positive Lebesgue measure.*

Proof of Theorem 2. By Theorem 1(i) applied to each group, vanishing of every summand gives $w \equiv b_j$ ρ_j -a.s. on U_j .

(i) Let $\{j, j'\}$ be an edge of $\mathcal{G}_c$ (Definition 1), so with $f_j = d\rho_j/d\mu$ the measure $d\nu = \min(f_j, f_{j'}) d\mu$ is nonzero. By construction $\nu \leq \rho_j$ and $\nu \leq \rho_{j'}$ as measures, hence $\nu \ll \rho_j$ and $\nu \ll \rho_{j'}$, so $w = b_j$ holds ν -a.s. and $w = b_{j'}$ holds ν -a.s. The union of the two ν -null exceptional sets is ν -null and $\nu \neq 0$, so there is a point at which both identities hold and $b_j = b_{j'}$. This is precisely the step that the weaker “some set charged by both measures” criterion does not license: two measures can each charge a common set while being mutually singular, and then no single point need carry both statements. Equality along edges propagates along paths, so $j \mapsto b_j$ is constant on each connected component.

(ii) If $\mathcal{G}_c$ is connected, all b_j equal a common $b(c)$, hence $w \equiv b(c)$ on $U_c = \bigcup_j U_j$ and $q_\theta(\mathbf{x} | c) = a(c) e^{-E(c, \mathbf{x})/kT}$ there, with $a(c) = e^{b(c)} > 0$. Theorem 1(ii) applies with $U = U_c$ and fixes the multiplier only relative to U_c : $a(c) = q_\theta(U_c | c)/Z_{U_c}(c)$, so the conditional law $q_\theta(\cdot | c, U_c)$ is exactly the Gibbs law conditioned on U_c . If in addition U_c carries all the model’s mass, $q_\theta(U_c | c) = 1$, then $a(c) = 1/Z_{U_c}(c)$; and only if U_c also carries all the target’s mass, so that $Z_{U_c}(c) = Z(c)$, does this become $q_\theta(\cdot | c) = \pi(\cdot | c)$ globally. Neither hypothesis holds in any run reported here (Remark 9 and corollary 3).

(iii) Work in the continuous reading, where by hypothesis each ρ_j has a positive Lebesgue density on U_j . Suppose $\mathcal{G}_c$ has components $\mathcal{S}_1, \dots, \mathcal{S}_C$ with regions $U^{(i)} = \bigcup_{j \in \mathcal{S}_i} U_j$. Two distinct components have Lebesgue-null intersection: take $\mu = \lambda$ Lebesgue measure in this reading, and suppose $U^{(i)} \cap U^{(i')}$ had positive Lebesgue measure. Then some pair $j \in \mathcal{S}_i, j' \in \mathcal{S}_{i'}$ has $\lambda(U_j \cap U_{j'}) > 0$, and on that common set f_j and $f_{j'}$ are both strictly positive by hypothesis, so $\min(f_j, f_{j'}) > 0$ there and $\int \min(f_j, f_{j'}) d\lambda > 0$, which is (16); Definition 1 would then place an edge between them, contradicting that they lie in different components. Lebesgue-nullity is what the normalisation decomposition below needs, which is why the positive-density hypothesis is carried explicitly into parts (ii) and (iii) rather than built into Definition 1: under the atomic reference measures our runs actually use, every U_j is Lebesgue-null and this decomposition is unavailable, which is why Corollary 3 states the weaker identifiability conclusion instead. By (i) the residual is constant on each $U^{(i)}$, say $w \equiv b_i$, so $q_\theta(\mathbf{x} | c) = a_i e^{-E(c, \mathbf{x})/kT}$ on $U^{(i)}$ with $a_i = e^{b_i} > 0$; conversely every such density lies in the zero-loss set of the objective, because each $\text{Var}_{\rho_j}[w]$ is

then a variance of a constant. (This is a statement about the objective’s null set, not an existence claim about the model class: we do not assert that a flow of the architecture used here realises every member of the family.) Normalisation over U_c is the single scalar constraint $\sum_i a_i Z^{(i)} = 1$ on the positive multipliers, leaving $C - 1$ free parameters, which is (17). Finally $q_\theta(U^{(i)} | c) = a_i Z^{(i)}$, and as $(a_1, \dots, a_C)$ ranges over the constraint set these occupancies range over the whole open simplex; the Gibbs occupancies *conditioned on* U_c , $Z^{(i)} / \sum_{i'} Z^{(i')}$, are one point of it and are singled out by nothing in the objective. Note that these are conditional on U_c : the objective says nothing about π ’s mass outside U_c either. $\square$

Proof of Corollary 2. Each parent contributes exactly one group, and the operative reference measure is the empirical measure on that group’s sampled geometries (Remark 8). The conclusion follows from mutual singularity rather than from any count of groups per composition. Two groups are joined in $\mathcal{G}_c$ exactly when their empirical measures are not mutually singular, which for atomic measures means exactly when the two clouds share a sampled geometry (Corollary 3(b)); distinct parents draw their displacements independently from continuous distributions, so that event has probability zero and the two empirical measures are almost surely mutually singular. Every vertex is therefore isolated whatever J is, each connected component is a single group, and Corollary 3 applies with one free offset per cloud. Theorem 1(i) in its atomic form is the strongest part available.

Two remarks keep this honest. First, the argument is deliberately independent of composition multiplicity, because the training pool does contain repeated compositions (Appendix B.4); those repeated clouds are still edgeless, so the conclusion is robust to multiplicity in a way that a $J = 1$ argument would not be. Second, the conclusion concerns the supervision design and not the geometry of Gaussian tails: were the ρ_j read as full-support Gaussians, $\mathcal{G}_c$ would be complete, but then the population objective would be a variance over all of $\mathbb{R}^{3N}$, which no run here approaches, and the *measured* certificate would still be the $K - 1$ empirical equations. Even for a connected $\mathcal{G}_c$ the normalisation hypothesis of Theorem 2(ii) is unattainable for a flow with full-support prior, so the reachable continuous statement is the ε -relaxed one of Remark 9. $\square$

Remark 9 (The normalisation hypothesis is an idealisation, and what survives without it). No continuous normalising flow driven from a full-support Gaussian prior satisfies $q_\theta(U_c | c) = 1$ for a bounded U_c , so parts (ii) of Theorems 1 and 2 are limits rather than attainable states, and any upgrade path through them must be read as approximate. The ε -relaxation is what actually survives: if $w \equiv b$ on U_c and $q_\theta(U_c | c) = 1 - \varepsilon$, then with the positive multiplier $a = e^b$ we have $a Z_{U_c}(c) = 1 - \varepsilon$, so $a = (1 - \varepsilon) / Z_{U_c}(c)$ and $b = -\log Z_{U_c}(c) + O(\varepsilon)$, and the law of q_θ *conditioned* on U_c is exactly the Gibbs law *conditioned* on U_c . Connectivity therefore buys a composition-level constant relative to the mass inside U_c , not a global normalisation.

Remark 10 (What would test the theorem). Theorem 2 is falsifiable: a model trained with $C > 1$ can have arbitrarily wrong mode occupancies while achieving zero training loss and a per-group correlation of 1. Testing it requires groups that are not mutually singular — which, under the empirical reference measures of Remark 8, means groups that share sampled geometries: perturbation clouds tiled along a transition path so that consecutive clouds re-use the same geometries, or groups drawn from one reference ensemble obtained by enhanced sampling. Merely intersecting supports would not suffice. That must be combined with a measurement of relative basin populations rather than within-basin ordering. No experiment in this paper does that (Section 5).

Remark 11 (Why cross-molecule pooling is broken). Across molecules the certificate holds with a *molecule-dependent* constant $-\log Z(c)$ whose spread is dominated by size and composition. A cross-batch variance therefore measures $\log Z$ spread rather than Boltzmann deviation. This is the same Simpson-type trap that forces the evaluation correlation to be computed per parent molecule (Section 4): one argument covers both the loss and the metric, and it is why the within-parent form is the only one used in either.

A.7 ALGEBRAIC RELATION BETWEEN RESIDUAL VARIANCE AND ORDERING

The objective penalises a within-group variance while the endpoint reports a within-group correlation. They share one residual form, which is why optimising the first is expected to move the second — under a stated condition that our own measurements show is not met.

Proposition 4 (Residual-variance / correlation identity). *Fix a parent with K evaluation geometries; let $u_k = -E_k/kT$ be the independent evaluation energy, $L_k = \log q_\theta(x_k)$, $w_k = L_k - u_k$, $\tau = \text{sd}_k(w)/\text{sd}_k(u)$ and $\rho = \text{corr}_k(w, u)$. Then the per-group Pearson correlation is $r = (1 + \rho\tau)/\sqrt{1 + 2\rho\tau + \tau^2}$, reducing at $\rho = 0$ to $r = 1/\sqrt{1 + \tau^2}$, and that parent’s contribution to (2) is exactly $\tau^2 \text{Var}_k(u)$.*

Proof of Proposition 4. Fix a parent m with K evaluation perturbations. Write $u_k = -E_k/kT$ for the independent evaluation energy, $L_k = \log q_\theta(x_k)$, and $w_k = L_k - u_k$ for the Boltzmann residual, and set $\tau = \text{sd}_k(w)/\text{sd}_k(u)$ and $\rho = \text{corr}_k(w, u)$. From $L = u + w$,

$$\text{Cov}(L, u) = \text{Var}(u) (1 + \rho\tau), \quad \text{Var}(L) = \text{Var}(u) (1 + 2\rho\tau + \tau^2), \quad (18)$$

so the per-group Pearson correlation is

$$r_m = \frac{1 + \rho\tau}{\sqrt{1 + 2\rho\tau + \tau^2}}, \quad \text{and, if } \rho = 0, \quad r_m = \frac{1}{\sqrt{1 + \tau^2}}. \quad (19)$$

At $\rho = 0$ the right-hand side is strictly decreasing in τ , and $\mathcal{L}_{\text{value}}$ for parent m is exactly $\tau^2 \text{Var}_k(u)$. Hence at $\rho = 0$ minimising the training loss minimises the residual that the reported metric penalises. Away from $\rho = 0$ the two can move in opposite directions, which is what Section 4.3 measures: the observed disagreement shows that the simplifying condition does not hold, without identifying the sign or the distribution of ρ (Remark 12). This proposition is a purely algebraic identity, not a guarantee that reducing the loss improves the correlation. $\square$

Remark 12 (τ^2 is the normalised residual variance). With $u_k = -E_k/kT$ we have $\text{Var}_k(u) = \text{Var}_k(E/kT)$, so the metric (4) satisfies

$$\text{NRV}_m = \frac{\text{Var}_k[w]}{\text{Var}_k[E/kT]} = \tau_m^2, \quad (20)$$

and by Proposition 4 parent m ’s contribution to (2) is $\text{NRV}_m \text{Var}_k(u)$. Within a single group under a single potential, then, NRV is exactly the τ^2 of Proposition 4, and the metric is the normalised evaluation analogue of the same Boltzmann-residual form the training objective uses. The two are not one quantity in different units: they share that residual structure but are evaluated with different potentials (eSEN against GFN2-xTB), different populations (training shards against held-out parents) and at different discretisation accuracies — the same target quantity, a different Euler bias. The identity is also the bridge between the ordering result and the calibration result, and it shows that the two cannot both be read through the $\rho = 0$ idealisation. At $\rho = 0$, $r = 1/\sqrt{1 + \tau^2}$ is strictly decreasing in τ , so the arm with the larger r must have the smaller NRV. The measurement is the opposite: value supervision has both the larger r (0.369 against 0.200) and the larger NRV (2.144 against 1.402). These two endpoints are aggregated differently — r is a mean over parents and NRV a median over parents (Appendix B.7) — so no per-parent ρ can be recovered by combining them. What the disagreement does establish is that the zero-correlation simplification is not an adequate explanation of the observed arm-level behaviour: under it the arm with the larger r must have the smaller NRV, and no aggregation convention makes both reported endpoints consistent with that. The sign and the distribution of the residual correlation are *not identified* by these aggregates, and we report neither.

Remark 13 (The correlation certifies shape, not temperature or calibration). Pearson r is invariant under positive affine maps of either variable, so it cannot distinguish $p \propto e^{-E/kT}$ from $p \propto e^{-E/kT'}$ for any $T' > 0$, and it is equally blind to the *magnitude* of the log-density’s response. The per-group regression slope, the scale ratio $\text{sd}_k[\log q_\theta]/\text{sd}_k[E/kT]$, the implied effective temperature kT/slope and the normalised residual variance (4) are the read-outs that are *not* affine-invariant, which is why they are reported alongside every correlation (Appendix B.7). Their verdict differs from the correlation’s, and that difference is the paper’s main empirical statement.

Remark 14 (Attenuation, and why we do not lean on it). By Assumption 4 the evaluated $\log q_\theta$ carries discretisation bias and stochastic trace noise. The classical attenuation model, $r_{\text{meas}} \approx r_{\text{true}}/\sqrt{1 + \text{Var}(\text{noise})/\text{Var}(\log p)}$, would make the reported correlation an under-estimate of the true one, and geometry-dependent discretisation bias would enter $\text{Var}_k(w)$ as an estimator floor. This is an argument and not a measurement, and we do not lean on it. What is measured instead is the teacher-evaluator agreement reference: pushing the teacher’s own energies through the identical pipeline gives $r = 0.927$ on the primary population, so a model reproducing its teacher exactly would

score about that rather than 1, and our 0.369 is about 40% of it in absolute terms (Appendix B.7). Whatever accounts for the residual, it is not teacher–evaluator disagreement.

A.8 SCOPE OF THE THEORY

Four inventories, so that no statement above is credited with more than it does.

Not ours. Theorem 1 and its partition-function-free property are the log-variance divergence of Richter et al. (2020); Nüsken & Richter (2021); Richter & Berner (2024), restated in our conditional notation; off-policy estimation of such a divergence is likewise established (Sendera et al., 2024). See Remark 6.

Exercised empirically. Theorem 1(i) in its atomic form, applied to each disjoint sampled cloud — that is the training objective; and Remark 11, which dictates the per-group construction of both the loss and the metric.

Stated as a diagnostic identity, with its hypothesis contradicted. Proposition 4 is exact algebra, but its $\rho = 0$ specialisation — under which reducing normalised residual variance must improve the correlation — is contradicted by the measurement of Section 4.3, where the value arm has both the larger r and the larger NRV (Remark 12). That failure is itself one of the paper’s findings.

Stated and not tested here. Theorem 1(ii) and Theorem 2(ii): no reported run has a connected overlap graph, so we never reach the regime in which the certificate becomes a statement about q_θ on a region — and even there it would pin only the law *conditioned* on that region (Remark 9). Theorem 2(iii) is the regime we *are* in, and its failure mode — arbitrary relative basin weights at zero loss — is untested in both directions, because no cross-basin measurement exists in this paper (Remark 10, Section 5). Propositions 2 and 3 are candidate mechanisms for the negative result of Section 4.1, consistent with it but not isolated experimentally.

Outside the theory altogether. Nothing above predicts the *magnitude* of the correlations we report, and nothing above speaks to the *calibration* of $\log q_\theta$: Theorem 1(i) fixes the residual only up to a per-group constant, and by Remark 13 the per-group correlation is invariant to positive affine maps of either variable. The scale-sensitive read-outs — normalised residual variance, regression slope, scale ratio and effective temperature — are therefore empirical additions with no counterpart here, and the finding that they fail to improve while the correlation does (Section 4.3) is not something the theory anticipates; the only bridge between the two is the bookkeeping identity (20). Finally, every statement above concerns population objectives at their exact optimum, whereas (2) is estimated from M parents with a discretised, stochastic $\log q_\theta$, and the update we run is the truncated-adjoint surrogate of Remark 1. Everything above is therefore a statement about the zero set of the ideal residual condition and about what that zero set leaves unidentified; nothing above is a statement about whether, or to what, the implemented update converges.

B IMPLEMENTATION AND EXPERIMENTAL DETAILS

This appendix is the reproducibility record. It states the objectives exactly as implemented rather than as displayed (Appendices B.1 and B.2), the training and evaluation protocols (Appendices B.3 to B.5), the two diagnostics that explain the gradient baseline’s behaviour, together with the score read-out those cells use (Appendix B.6), the full per-seed tables behind each reported mean (Appendices B.7 and B.9), the two experimental rounds and the defects of the earlier one (Appendices B.9 and B.10), the auxiliary measurements (Appendices B.8, B.13 and B.14), the numerical-stability record (Appendix B.15), the complete limitations inventory (Appendix B.17), and the reproduction path (Appendix B.19).

B.1 THE OBJECTIVES AS IMPLEMENTED

The implemented objectives deviate from the displayed ones in ways that are material, and are listed here rather than left to a reader of the code.

Clipping and Huberisation. The score read-out of (3) is norm-capped at 1000 per atom before the residual against the teacher force is formed, and the per-atom squared error is Huberised above 10^4 ; the anchor term clamps $\log q_\theta$ at $\pm 10^6$; and a value of $\mathcal{L}_{\text{value}}$ above the value-loss cap is rejected outright for that step rather than clipped, because such values are finite and therefore invisible to a non-finiteness test. The implemented $\mathcal{L}_{\text{force}}$ and $\mathcal{L}_{\text{value}}$ are accordingly clipped and Huberised variants of the stated objectives. All thresholds are in Table A1; for a converged model they are inactive.

Frozen-trajectory gradient. The reverse-time trajectory used for the continuous-flow log-density is advanced without gradient tracking, each state is detached before its divergence is taken, and the prior term is computed from a detached endpoint. The gradient actually back-propagated through $\mathcal{L}_{\text{value}}$ is therefore

$$\nabla_\theta \mathcal{L}_{\text{value}} \approx \sum_{\text{steps}} \Delta t \nabla_\theta (\nabla \cdot v_\theta(\mathbf{x}_{\text{step}}, t_{\text{step}})) \quad (\text{trajectory held fixed}), \quad (21)$$

a truncated-adjoint surrogate: the reverse trajectory and the prior endpoint are held fixed and only the divergence evaluations are differentiated. We do not differentiate through the ODE solve, and what the optimiser performs is therefore not exact likelihood-gradient descent — neither on the continuous-time likelihood nor on its discretisation — but descent on this surrogate.

Three scopes are kept apart throughout the paper, and this paragraph is the place they meet. *Theory*: the identifiability results of Appendix A characterise the zero set of the displayed ideal residual objective — what a model that had driven the population objective to zero would have been shown, and what that leaves free — and make no claim about whether the implemented update converges to that zero set (Remark 1). *Implementation*: what runs is the finite-step density estimate of Appendix A.2 optimised by the truncated-adjoint gradient displayed above, together with the clipping, Huberisation and step rejection listed in this subsection. *Experiment*: every empirical result in this paper is a measurement of the behaviour of that implemented procedure, and none of it is offered as evidence about convergence to the ideal zero set. Where a statement belongs to one scope rather than another we say so at the statement.

Which code path is live, and what the reported model is. The reported value term is the *within-parent* variance of the Boltzmann residual. A cross-batch variance form also exists in the codebase; it is retained only as a deprecated reference for unit tests and is mathematically inappropriate here (Remark 11). In every result reported as ours, the model is the two-term configuration $\lambda_1 = 0$, $\lambda_3 = 0$, $\lambda_E = 3 \times 10^{-5}$: flow matching plus the value term. The gradient term is defined in Section 2.5 because Section 4.1 measures it, not because any reported model uses it; wherever a result comes from a configuration with $\lambda_1 > 0$ we say so at that result.

The inactive anchor code path. A third term, $\mathcal{L}_{\text{anchor}}$, exists in the implementation and was inactive ($\lambda_3 = 0$) in *every* run reported here. It appears in no equation of Section 2 and has no experimental result in this paper; the record is kept because the code path ships, and because it should not be mistaken for an untried repair of Section 4.3. It would constrain the additive group offset that (2) cannot see (Corollary 1), and NRV is invariant to exactly that offset, so no setting of λ_3 could move the calibration endpoint. Its implemented form, $\text{mean}[(\log q_\theta + E/kT + \log \hat{Z})^2]$, decomposes into the same within-group variance the value term already targets plus a squared offset penalty and introduces no read-out sensitive to the *magnitude* of the density response; and its predictor $\log \hat{Z}$ reads element counts, atom count and total charge only, so it can fix at most one constant per composition rather than the one per group Corollary 1 identifies.

The value estimator as implemented. Equation (2) displays $M^{-1} \sum_m \text{Var}_k(w_{m,k})$. The implementation uses the biased (divide-by- K) variance, drops non-finite residuals, excludes any group left with fewer than two valid members, and averages over the surviving groups rather than over M ; separately, a whole step’s value loss is rejected when it exceeds the cap above, so the optimised objective is a conditionally sampled version of the displayed one. The null-space statements of Theorem 1 and corollary 1 are unaffected by the biased normalisation — a variance is annihilated by the same functions either way — but they are stated for the full-group objective.

Table A1: Constants that are part of the definition of the implemented objectives and of the evaluation protocol. The clipping thresholds are inactive for a converged model but are part of the objective as trained, so they are stated here rather than left to the code. Where two settings are listed, “earlier round” names the checkpoint family of Appendix B.10 and “matched grid” the family of Appendix B.9; the names are family labels and carry no claim that either configuration achieved numerical stability, which Appendix B.15 reports it did not.

Symbol / name	Value	Where it acts	Role
<i>Objective</i>			
λ_1	0.1 (gradient cells), 0 otherwise	$\mathcal{L}_{\text{force}}$	weight of the gradient-matching term
λ_E (code name: lambda_2)	3×10^{-5} (value cells), 0 otherwise	$\mathcal{L}_{\text{value}}$	weight of the within-parent Boltzmann-residual variance
λ_3	0 in every reported run	$\mathcal{L}_{\text{anchor}}$	gauge-fixing anchor, inactive
kT	1.0 eV ($\approx 11\,600$ K)	both physics terms	temperature scale, not physical
σ	1.0	prior, read-out	prior standard deviation
probe times t	0.85, 0.92, 0.97	$\mathcal{L}_{\text{force}}$	amplification 5.7/11.5/32.3 $\times$
warm-up / ramp	1% / 5% of training	schedule	FM only, then a linear ramp
<i>Clipping: the objectives as implemented</i>			
score norm cap	1000 per atom	s_θ	caps the read-out
per-atom error cap	10^4	$\mathcal{L}_{\text{force}}$	Huber transition
$\log q_\theta$ clamp	10^6	$\mathcal{L}_{\text{anchor}}$	inactive ($\lambda_3 = 0$)
value loss cap	3000 (earlier round), 1500 (matched grid)	$\mathcal{L}_{\text{value}}$	rejects finite outliers
<i>Density estimator</i>			
ODE steps	4	FFJORD	explicit Euler in state, midpoint-rule divergence quadrature
Hutchinson probes	2	divergence	Rademacher
parents per step M	4 (earlier round), 8 (matched grid); 1 diagnosed only	$\mathcal{L}_{\text{value}}$	estimator variance $\propto 1/M$
step interval	every 4th step	$\mathcal{L}_{\text{value}}$	amortises the FFJORD cost
atoms per parent	≤ 50	$\mathcal{L}_{\text{value}}$	memory bound
<i>Data and evaluation</i>			
max atoms	200	preprocessing	molecule size cut
elements	83	preprocessing	bond-free, no bond labels
group size K	5 train (ref. + 4) / 9 eval (ref. + 8)	shards, eval	within-parent groups
shard σ	0.03–0.40 Å (5 values)	shards	displacement scales
eval σ	0.15 Å	eval	perturbation scale
parents per arm	120 (30 early round)	eval	evaluation groups

B.2 HYPERPARAMETERS AND NUMERICAL CONSTANTS

The physics weights are $\lambda_E = 3 \times 10^{-5}$ in the value cells and $\lambda_1 = 0.1$ in the gradient-supervision cells; $\lambda_3 = 0$ everywhere. λ_E was fixed once from a smoke run, before any density–energy evaluation, and never tuned against the reported metric: on that run $\mathcal{L}_{\text{value}} \approx 6060$ against $\mathcal{L}_{\text{FM}} \approx 0.25$, so the weight makes $\lambda_E \mathcal{L}_{\text{value}} \approx 0.18$, the same order as the flow-matching term, whereas a ten-fold larger weight would have contributed ≈ 3.0 and swamped it. The weight is small; the term’s *contribution* is not, and we use “lightly weighted” for the former only.

The temperature scale is $kT = 1.0$ eV ($\approx 11\,600$ K) in every reported run: it is a numerical choice that keeps E/kT and F/kT tame, not a physical temperature. Because the variance-form objective is scale-covariant rather than scale-invariant, the choice of kT fixes the *target within-group slope* of log-density against energy that the objective asks the model to match; it does not imply that the trained model represents the Boltzmann distribution over configuration space as a whole. The score is probed at $t \in \{0.85, 0.92, 0.97\}$ during training; the prior standard deviation is $\sigma = 1.0$. Physics terms are switched on after a short flow-matching-only warm-up (1% of training) and ramped linearly to full weight over the following 5%. The value term is evaluated every fourth optimiser step on parents of at most 50 atoms, which is what makes an FFJORD-based objective affordable at all.

B.3 TRAINING SCALE AND HARDWARE

Every reported run trains a single bond-free SE(3)-equivariant vector field (256 hidden scalar channels, 32 vector channels, 6 molecule-update blocks, 32-dim radial basis to 10 Å, self-conditioning enabled) on one 141 GB data-centre GPU with 256 GB of host memory and 16 data-loading workers, using AdamW at base learning rate 2×10^{-4} , weight decay 10^{-12} , a 2% linear warm-up, gradient-norm clipping at 1.0, mixed bfloat16 precision, batch size 8 with two gradient-accumulation steps, for 30 000 optimiser steps. Jobs run on a preemptible partition with a three-day wall clock and resume from the latest checkpoint on signal. Across the matched grid the cells differ only in the physics block of the configuration and in the random seed, which is what makes Appendix B.9 architecture-matched. The earlier checkpoint family is matched in architecture, data pipeline, optimiser, schedule and step budget, but not in parents per value step, loss cap or shard list (Appendix B.10).

B.4 DATA, PREPROCESSING, AND THE PERTURBATION SHARDS

Training data is the OMol25 4M split (Levine et al., 2025): 3.9M molecules, 205M atoms, 83 elements, with DFT forces and energies and *no* bond labels. Preprocessing converts the raw records into flow-matching graphs with positions, atom-type indices stored as `int8`, per-atom formal charges, and per-atom force and per-molecule energy labels read through the standard single-point-calculator interface. Molecules with more than 200 atoms are excluded. Bond tensors are written empty and the bond loss weight is zero, so no bond supervision enters training at any point; where bonds are needed for evaluation only, they are recovered post hoc from the generated geometry with `xyz2mol` (Kim & Kim, 2015).

The value term consumes pre-computed perturbation shards rather than the main data loader. Each shard stores, for each parent molecule, $K = 5$ geometries — the data geometry plus four displacements at standard deviations $\{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å — together with their teacher energies, in K -contiguous blocks with a parent identifier, so that a within-parent variance can be formed by reading K consecutive entries. Note that a group contains the data geometry *together with* its $K - 1$ displacements; the primary endpoint drops the data geometry at evaluation only, never in training.

Composition multiplicity, and why Corollary 2 does not rest on it. Reading the composition of each parent as (element multiset, total charge) from the stored per-atom types and charges, the 30 000-parent training shard carries only 22 822 distinct compositions: 2 204 compositions are shared by more than one parent, covering 9 382 parents (31%), at multiplicity up to 80. It would therefore be false to describe the supervision design as contributing one group per composition. What is true, and what Corollary 2 uses, is that each parent supplies one empirical perturbation cloud and that clouds sharing a composition almost surely share no sampled geometry, so the empirical overlap graph is edgeless whatever the multiplicity. In the evaluated population — the 120 grid parents and the 93 primary parents — every parent does carry a distinct composition, but no step of the argument uses that.

The shard audit. Two defects in the shard configuration of the earlier round are the reason the primary endpoint is defined the way it is, and the reason the matched grid of Appendix B.9 was built. Both are stated once, here.

(a) *One shard came from the split the evaluation parents are drawn from.* Every value arm of the earlier round reads two shards: a 30 000-parent shard built from the training split and a 10 000-parent shard built from the validation split. The density evaluation draws its held-out parents from that same validation split, so roughly a quarter of the evaluation parents were inside the value term’s training pool, with their true teacher energies, while the flow-matching control — which disables the physics block and reads no shard — was unaffected: the overlap is arm-specific and favours the arm that wins. Matching each evaluation parent’s data geometry back to the validation file by a translation-invariant signature identifies 118 of 120 parents, 25 of them inside the pool; the remaining 93 are the disjoint set on which the primary endpoint of Table A4 is computed. The correct repair is to rebuild both shards from the training split and re-run the arms, which is what the matched grid does: every cell of Appendix B.9 reads training-split shards only.

(b) *The earlier scrambled control was scrambled in only one of its two shards.* The y -randomisation control (Rücker et al., 2007) is produced from the training-split shard by permuting the geometry-

to-energy assignment *within* each parent, so that the energy multiset and the within-parent energy standard deviation (77.6 kcal/mol) are bit-for-bit unchanged and only the pairing differs. No scrambled validation shard was ever built, and that arm’s configuration lists the unscrambled validation shard alongside the scrambled one, so about a quarter of its parent pool carried true pairings: the negative control was partly a positive control. A second, smaller defect is that its three seeds are not all configuration-matched to the true-label arm — two use $M = 4$ and cap 3000, matching all four true-label seeds, and the third uses $M = 8$ and cap 1500 with batch 4. Cell C of Appendix B.9 repairs both. The general lesson is worth one sentence, because the failure is silent and reproducible: a y -randomisation control is only a control if *every* input it reads is randomised, and a shard list is exactly the kind of configuration surface on which that invariant fails without any error being raised.

B.5 EVALUATION PROTOCOL

Local density–energy ordering is measured on held-out parent molecules. The Stage-1 pipeline emits $K = 9$ geometries per parent: its data geometry, plus 8 copies displaced with per-coordinate standard deviation $\sigma = 0.15 \text{ \AA}$ (centre of mass removed). We compute $\log q_\theta$ for each by an ODE integration of the clamped positional flow with Hutchinson divergence estimation (Appendix A.2). The data geometry is systematically its group’s lowest-energy member and therefore high-leverage, so the primary endpoint discards it and scores the 8 displaced geometries alone.

Because the evaluation pipeline aggregates to per-group correlations and discards the individual $(\log q_\theta, E)$ pairs, we re-scored every geometry in every arm and recompute every reported statistic from that per-record source, composing the group-to-validation-index map of the contamination check with the reference-exclusion logic. The re-scoring loses 6 of 1080 single points per arm rather than the pipeline’s 1–2, which shifts a mean by at most 0.004. Since every reported number comes from the single re-scored source, they are mutually comparable.

Each geometry is then scored with GFN2-xTB (Bannwarth et al., 2019), a semi-empirical method *independent* of the neural potential used for training: it enters no training loss and shares no parameters, data or functional form with the teacher. Of the 1080 attempted geometries per arm, 1078–1080 were scored; the one or two single points that failed to converge were dropped from their group before the correlation was formed. We report the Pearson correlation between $\log q_\theta$ and $-E_{\text{GFN2-xTB}}/kT$ computed *within* each parent and then averaged over parents. Grouping is not a stylistic choice: pooling all parents into one regression gives $r^2 = 0.11$ for the value arm and 0.01 for the flow-matching control, because between-molecule $\log Z$ offsets dominate the pooled variance (Remark 11).

How comparisons are reported. The unit of replication is a training run. We report effect sizes, means and standard errors over independently trained seeds, together with every seed-level result. Because several cells contain only two to five completed seeds and completion is objective-dependent (Appendix B.15), we use these descriptive quantities rather than treating asymptotic null-hypothesis tests as the primary evidence. Every arm-level number in this paper is therefore an arm mean $\pm$ the standard error over seeds, quoted with the number of seeds behind it and, wherever completion differs by arm, with the launched, completed and diverged counts; every comparison is quoted as the absolute difference Δr of two such means and read against the observed seed ranges, which the per-seed tables print in full. The inferential statistics computed during the analysis are collected once, for completeness and in one place, in Table A16; no claim in this paper is stated in terms of them.

Five seed configurations were prepared for the flow-matching and value arms and seven for the scrambled arm, of which five, four and three are reported, and the reasons differ by arm rather than sharing one cause.

All five flow-matching seeds completed and all five are scored, the fifth having carried no score under this protocol until it was scored under it (Appendix B.10). The unreported value run and two of the unreported scrambled runs diverged to non-finite weights before the training budget was reached and were not replaced; a further scrambled run diverged after its mid-budget checkpoint, which is not the reported budget and is not scored; and one scrambled run completed with finite weights but lost its evaluation to a launcher path fault and has since been recovered (Appendix B.10).

Adverse imputation for the unreported run. The adverse imputation charges the value arm alone, the control arm having no unreported run left to impute. On the primary endpoint the value arm’s four completed seeds score 0.316, 0.378, 0.426 and 0.358 and the control arm’s five score 0.222, 0.194, 0.249, 0.142 and 0.192, giving arm means 0.369 and 0.200 and an observed difference of $\Delta r = +0.169$ (Table A5). Two imputations replace the value arm’s one diverged run with a value chosen against us and recompute that arm’s mean over five entries rather than four. (i) Imputed at the control arm’s own mean, 0.200, the value arm falls to 0.336 and the difference to $\Delta r = +0.135$. (ii) Imputed at the worst value observed in *any* arm of the round, 0.142 — the control arm’s fourth seed, and by construction the most adverse single value available, since no arm of the round produced a lower one — the value arm falls to 0.324 and the difference to $\Delta r = +0.124$. Under both imputations the imputed five-entry value-arm mean still lies above every one of the five control seeds, whose maximum is 0.249. The two conservative imputations are reported here and referenced from Section 4.2.

B.6 WHY THE GRADIENT BASELINE UNDERPERFORMS: TWO DIAGNOSTICS

What Section 4.1 measures to fail is naive off-policy endpoint force matching, read through the flow-matching score identity (3), on this non-equilibrium corpus. This subsection reports two diagnostics that supply a *measured mechanism* for that failure. Both were run on the shared warm-start flow-matching checkpoint in forward and backward passes only: nothing was trained, no checkpoint was written, and no number reported anywhere else in this paper depends on them. They are offered as a mechanism and not as a proof, and they do not widen the claim, which remains scoped exactly as stated in Section 2.5: naive, off-policy, endpoint, read through the flow-matching score identity, on this non-equilibrium corpus.

Diagnostic 1: the endpoint force is the wrong target at the interpolant. The gradient term regresses the read-out at the interpolation state x_t onto the force read at the data endpoint x_1 . Table A2 measures how far apart those two forces actually are. It sweeps the probe time over 192 held-out molecules with the *same* x_0 and x_1 at every probe time, so the sweep is paired and a difference between rows is a difference in probe time alone. For each row it reports the per-atom mismatch $\|F(x_t) - F(x_1)\|$, the per-atom cosine between the two forces, that mismatch relative to the magnitude of the endpoint force it is supposed to match, and the norm of the parameter gradient the force loss produces.

Table A2: The endpoint force against the force at the interpolant, 192 held-out molecules, paired across rows: the same x_0 and the same x_1 are used at every probe time, so the only configured variable down the table is t . Columns are the per-atom mismatch $\|F(x_t) - F(x_1)\|$, the per-atom cosine between the two forces, the mismatch divided by the endpoint force magnitude, and the norm of the parameter gradient of the force loss. The endpoint force itself averages 1.49 eV/Å over this population. Measured on the shared warm-start flow-matching checkpoint in forward and backward passes only; no training.

probe time t	mismatch (eV/Å)	$\cos(F(x_t), F(x_1))$	mismatch / $\ F(x_1)\ $	$\ \nabla_{\theta} \mathcal{L}_{\text{force}}\ $
0.80	10.66	0.12	17.9	21.5
0.85	7.27	0.16	12.2	25.1
0.90	4.33	0.24	7.1	48
0.92	3.30	0.29	5.4	57.9
0.95	1.93	0.40	3.1	94
0.97	1.11	0.52	1.8	150

The endpoint force averages 1.49 eV/Å on this population against a mismatch of 10.66 eV/Å, so on population averages the mismatch is about seven times the target it is supposed to match. Taken per molecule, each mismatch against that molecule’s own endpoint force, the ratio averages 17.9: molecules whose endpoint forces sit near zero contribute large ratios and the mean of ratios exceeds the ratio of means. On either reading the target is nearly orthogonal to what is being matched, at a per-atom cosine of 0.12. The reason is visible in the same run: $\|F(x_t)\|$ averages 10.98 eV/Å there, because the interpolant sits off the data manifold, where atoms are close and forces are large. Moving the probe later reduces the mismatch but raises the gradient norm sevenfold. Both ends of the sweep are bad: early probes match a target that barely relates to the endpoint force, and late probes are dominated by gradient magnitude. This is the trade-off Proposition 2 names, now measured rather than argued.

That the mismatch is a property of the *location* and not of the teacher is checked in the same run. The teacher’s force at $\mathbf{x}_1$ agrees with the dataset’s own labels to a per-atom cosine of 0.956, with magnitudes 1.4878 against 1.4901 eV/Å. The teacher is therefore reproducing the corpus’s forces at the data geometries; what it does not reproduce is the endpoint force at a point that is not the endpoint.

Diagnostic 2: the force update opposes the flow-matching update; the value update does not. Table A3 reports, per minibatch, the cosine between the flow-matching parameter gradient and each of the two physics parameter gradients, computed at the same probe times.

Table A3: Cosine between the flow-matching parameter gradient and each physics parameter gradient, per minibatch, on the shared warm-start flow-matching checkpoint. Forward and backward passes only; no training and no parameter update was applied.

probe time t	$\cos(g_{\text{FM}}, g_{\text{force}})$	$\cos(g_{\text{FM}}, g_{\text{value}})$
0.80	−0.523	+0.116
0.85	−0.509	+0.072
0.90	−0.347	+0.024
0.92	−0.446	−0.000
0.95	−0.259	−0.032
0.97	+0.055	−0.045

Measured instead against the flow-matching gradient at its own randomly drawn time, the force cosine runs −0.174 to −0.084 and the value cosine is −0.009.

The reading. Both terms inject physical information into the same parameters, and the two diagnostics say they do it differently. The force update points systematically *against* flow matching over most of the probe range, while the value update is close to orthogonal to it everywhere and never opposes it by more than the force term does at its mildest. Read with Table A2, the picture is that the force term is regressing the network towards a target that is nearly orthogonal to the quantity it is evaluated at, and is pulling the parameters away from the flow-matching solution while doing so. We offer this as a measured mechanism for the difference reported in Section 4.1 and not as a proof of it: an anticorrelated gradient at a warm-start checkpoint constrains the first steps of an update, not the fixed point a full training run reaches, and neither diagnostic isolates a cause. Nothing here bears on force supervision in general (Guo et al., 2025; Hu, 2026), and nothing here licenses a broader claim than the one Section 2.5 makes.

The score read-out these cells use. The force cells are specified by one further detail, which we state exactly because the scope of Section 2.5 depends on it. Under the parameterisation this implementation uses, the network’s position output is the *endpoint prediction* $\hat{\mathbf{x}}_1$ rather than a velocity, and the gradient-supervision term passes that output to the score read-out (3) / Lemma 1. Because Lemma 1 maps a velocity to a score through a factor $t/((1-t)\sigma^2)$ while an endpoint prediction enters (12) through $t/((1-t)^2\sigma^2)$, the score these cells match is $(1-t)$ times the one (3) defines. This applies to every force number reported here: cells E and E2 of the matched grid, the force cells of the earlier round, and both diagnostics above.

What the diagnostics say about this scaling is that it sets the magnitude of the force term and not its direction. Both diagnostics were run under both read-outs and agree. Moving to the unscaled read-out changes the force loss by more than two orders of magnitude — at the latest probe time it rises from about 10 to about 4400 — while leaving the gradient cosines essentially unchanged: at the earliest probe time the cosine against the flow-matching gradient is −0.523 under the read-out these cells use and −0.509 under the unscaled one. The scaling therefore does not turn an anticorrelated update into an aligned one at the checkpoint probed. It does set the effective weight the force term carries, so a run under the unscaled read-out is a different configuration rather than the same one re-expressed, and Section 4.1 is scoped to the objective these cells ran.

B.7 CALIBRATION DIAGNOSTICS, PER-SEED VALUES, AND THE TEACHER-EVALUATOR AGREEMENT REFERENCE

This subsection gives the definitions behind the calibration endpoint, every per-seed value behind every mean in Table A4 and Figure 4, the sensitivity of those numbers to the two evaluation corrections, and how the teacher-evaluator agreement reference is measured.

Table A4: **The primary endpoint:** 93 held-out parents disjoint from the value term’s shard pool, data geometry dropped, 8 displaced geometries each, every energy a GFN2-xTB single point from an evaluator independent of the eSEN training teacher. r is the per-parent correlation between $\log q_\theta$, the clamped positional-flow density, and $-E/kT$, averaged over parents then seeds ($\pm$ SEM over seeds); NRV (4) is the median over parents per seed, then the mean over seeds, and is anchored: **a constant log-density scores exactly 1**. kT_{eff} ideally equals the 1.0 eV training target, and retrieval is against 12.5% and 37.5% at chance. The last row is *not a model* but the teacher’s own energies through the identical pipeline, reported for r and retrieval only; it is a reference for what exact reproduction of the teacher would score on this ensemble, not an attainable maximum, and its calibration diagnostics are derived in Figure 4. [†]The scrambled row is a defective control, one of its two shards having been left unscrambled and its third seed not configuration-matched, and is superseded by cell C of Figure 2; a fourth seed of that arm completed but lost its evaluation to a launcher fault and is recovered in Appendix B.10, where it lowers the arm further. Launch record for the three trained arms (Appendix B.10): flow matching 5 prepared / 5 completed / 0 diverged, all five scored here, the fifth having been a finite completed checkpoint that was scored under this protocol only later (Appendix B.10); value 5 / 4 / 1, all four scored; scrambled 7 / 4 / 3, three of them scored. Value entries are therefore conditional on completion (Section 2).

Arm	seeds	$r \uparrow$	NRV $\downarrow$	kT_{eff} (eV)	top-1 / top-3 $\uparrow$
Flow matching only	5	0.200 ± 0.018	1.402	8.02	20.2% / 49.5%
Value (ours)	4	0.369 ± 0.023	2.144	2.34	29.6% / 62.4%
Energies scrambled [†]	3	0.226 ± 0.020	n/a	n/a	n/a
<i>Teacher-evaluator agreement</i>	n/a	0.927	n/a	n/a	87.1% / 98.9%

Definitions, and the two identities that make them readable. For a parent with K geometries, NRV is (4), the slope is the ordinary least-squares slope of $\log q_\theta$ on $-E/kT$, the scale ratio is $\text{sd}_k[\log q_\theta] / \text{sd}_k[E/kT]$, and $kT_{\text{eff}} = kT / \text{slope} = -\text{Var}_k(E) / \text{Cov}_k(\log q_\theta, E)$, which is invariant to the kT used to form it and is therefore a statement about the model and not about our choice of scale. Writing $a = \text{Var}(\log q_\theta) / \text{Var}(E)$ and $b = \text{Cov}(\log q_\theta, E) / \text{Var}(E)$, rescaling $\log q_\theta \mapsto s \log q_\theta$ traces the parabola $\text{NRV}(s) = a s^2 + 2b s + 1$, minimised at $s^* = -b/a$ with value $1 - r^2$, which is the $\text{NRV}_{\min}$ of (4): no rescaling, and equivalently no single temperature, can bring the normalised residual below that floor. (We write the free parameter as the dimensionless rescaling s rather than as a temperature; the temperature reading $kT' = kT/s$ is a corollary.) Finally slope $= r \times$ (scale ratio) splits the slope into direction and magnitude, and by (20) NRV is exactly the τ^2 of Proposition 4. That identity holds within a single group under a single potential; NRV is therefore the normalised evaluation analogue of the same Boltzmann-residual form the training objective uses, and the two share that residual structure while being evaluated with different potentials and populations.

Aggregation, and why we quote medians. NRV is heavy-tailed across parents: groups whose $\text{Var}_k[E]$ is near zero divide by almost nothing, and one flow-matching seed has a *mean* NRV in the thousands driven by a single group. We therefore report the median over parents for each seed and then the mean over seeds, while r is a mean over parents and then over seeds; the analysis script additionally emits a 10% trimmed mean and bootstrap intervals. Paired per-parent comparisons are reported as the median paired shift in the per-parent, seed-median values over the same parents, together with the count of parents that move in each direction; the signed-rank statistics computed alongside them are collected in Table A16 and are not used anywhere in the argument.

Sensitivity to the two evaluation corrections. The primary endpoint applies two corrections at once — restrict to the 93 parents outside the shard pool, and drop the high-leverage data geometry — and both *reduce* the contrast we report. Table A6 gives the same diagnostics on the three other populations that the same records support, so that the sensitivity of every number to those two choices is visible. All four blocks come from one round, one set of seeds and one re-scored per-record source, and differ only in which parents and which geometries enter each group; none of them may be compared with the earlier, smaller round of Appendix B.10. The ordering endpoint is highest and

Table A5: Every seed behind Table A4 and Figure 4, on the primary endpoint (93 parents disjoint from the value term’s shard pool, data geometry dropped, GFN2-xTB energies). r and the scale ratio are means over parents within a seed; NRV, $\text{NRV}_{\min}$, the slope and kT_{eff} are medians over parents within a seed. Arm rows are the mean over seeds with the standard error over seeds. Recomputed for this table from the per-seed blocks of the calibration analysis, restricted to the 93 disjoint parents; the r and NRV columns reproduce Table A4 exactly. Seed numbers are within-arm labels listed in run order and do not pair across arms: the arms were trained independently and nothing here is a paired comparison at the seed level. *Seed 5 of the flow-matching arm is the later-scored control seed of Appendix B.10: a checkpoint that completed the full budget with finite weights but carried no score under this protocol until it was scored under it, on the identical parents, displacements and estimator settings.

Arm	seed	r	NRV	$\text{NRV}_{\min}$	slope	scale ratio	kT_{eff} (eV)
Flow matching only	1	0.222	1.393	0.810	0.142	0.906	7.04
	2	0.194	1.342	0.848	0.094	0.808	10.63
	3	0.249	1.337	0.816	0.162	0.854	6.17
	4	0.142	1.548	0.847	0.110	0.768	9.06
	5*	0.192	1.388	0.803	0.139	0.810	7.18
	<i>mean</i>	0.200	1.402	0.825	0.130	0.829	8.02
	<i>s.e.m.</i>	0.018	0.038	0.009	0.012	0.024	0.81
Value (ours)	1	0.316	2.955	0.782	0.427	1.763	2.34
	2	0.378	2.738	0.753	0.538	1.609	1.86
	3	0.426	1.547	0.714	0.513	1.210	1.95
	4	0.358	1.337	0.737	0.313	0.994	3.19
	<i>mean</i>	0.369	2.144	0.747	0.448	1.394	2.34
	<i>s.e.m.</i>	0.023	0.410	0.014	0.051	0.177	0.30
Energies scrambled	1	0.226	1.727	0.791	0.244	1.086	4.09
	2	0.192	1.757	0.829	0.176	1.001	5.70
	3	0.260	1.754	0.792	0.228	1.076	4.39
	<i>mean</i>	0.226	1.746	0.804	0.216	1.054	4.73
	<i>s.e.m.</i>	0.020	0.010	0.013	0.021	0.027	0.49

the calibration endpoint most flattering on the permissive population, which is why the restrictive one is primary.

Table A6: Sensitivity of every calibration diagnostic to the two evaluation corrections. Block 1 is the primary endpoint and is identical to the arm rows of Table A5; the other three blocks relax one or both corrections. Entries are means over seeds $\pm$ the standard error over seeds, with the same median-over-parents convention. Recomputed for this table from the per-seed blocks of the calibration analysis. A *constant* log-density scores $\text{NRV} = 1$ exactly, so no arm in any block is calibrated.

Arm	seeds	NRV	$\text{NRV}_{\min}$	slope	scale ratio	kT_{eff} (eV)	r
<i>Block 1 — 93 disjoint parents, data geometry dropped (the primary endpoint)</i>							
Flow matching only	5	1.402 ± 0.038	0.825 ± 0.009	0.130 ± 0.012	0.829 ± 0.024	8.02 ± 0.81	0.200 ± 0.018
Energies scrambled	3	1.746 ± 0.010	0.804 ± 0.013	0.216 ± 0.021	1.054 ± 0.027	4.73 ± 0.49	0.226 ± 0.020
Value (ours)	4	2.144 ± 0.410	0.747 ± 0.014	0.448 ± 0.051	1.394 ± 0.177	2.34 ± 0.30	0.369 ± 0.023
<i>Block 2 — all 120 parents, data geometry dropped</i>							
Flow matching only	5	1.382 ± 0.029	0.827 ± 0.009	0.148 ± 0.010	0.833 ± 0.019	6.88 ± 0.46	0.215 ± 0.015
Energies scrambled	3	1.691 ± 0.015	0.797 ± 0.012	0.236 ± 0.027	1.084 ± 0.034	4.35 ± 0.51	0.254 ± 0.022
Value (ours)	4	2.024 ± 0.377	0.736 ± 0.015	0.548 ± 0.053	1.420 ± 0.183	1.88 ± 0.19	0.397 ± 0.022
<i>Block 3 — 93 disjoint parents, data geometry retained</i>							
Flow matching only	5	1.303 ± 0.029	0.849 ± 0.004	0.028 ± 0.008	0.514 ± 0.005	52.2 ± 15.9	0.060 ± 0.016
Energies scrambled	3	1.061 ± 0.051	0.782 ± 0.020	0.232 ± 0.023	0.723 ± 0.023	4.40 ± 0.41	0.325 ± 0.037
Value (ours)	4	1.189 ± 0.173	0.711 ± 0.030	0.451 ± 0.046	1.049 ± 0.144	2.30 ± 0.28	0.420 ± 0.034
<i>Block 4 — all 120 parents, data geometry retained (the fully permissive metric)</i>							
Flow matching only	5	1.282 ± 0.027	0.845 ± 0.005	0.059 ± 0.007	0.556 ± 0.009	17.8 ± 2.1	0.088 ± 0.014
Energies scrambled	3	1.062 ± 0.045	0.777 ± 0.021	0.259 ± 0.033	0.736 ± 0.029	3.97 ± 0.46	0.341 ± 0.039
Value (ours)	4	1.211 ± 0.181	0.708 ± 0.030	0.491 ± 0.045	1.093 ± 0.159	2.09 ± 0.21	0.434 ± 0.032

The paired per-parent comparisons in full. The seed-level tests above have four and five units of replication, which is the unit of replication for a claim about a training method. The parent-level tests below have 93 units *within* those same nine models: they show how the shift is distributed across the evaluation set, and they are not additional training replications. All are paired per-parent differences of seed-median values over the same 93 parents with the data geometry dropped, value

arm minus flow-matching control, recomputed for this pass from the per-seed calibration blocks; we quote the median paired shift, and the value term improves on every ordering-flavoured diagnostic: per-group correlation (median $\Delta = +0.182$), regression slope ($+0.346$), scale ratio ($+0.500$) and the temperature-optimal bound $1 - r^2$ (-0.055). On NRV it is *worse*, by $+0.476$. The value term raises the per-parent correlation on 67 of the 93 parents. Parent-level paired analyses therefore show the shift distributed broadly across the evaluation set, and the two endpoints disagree in sign there as well as at seed level, which is the finding Section 4.3 reports and limitation (i) records. The fraction of parents clearing the constant-density bar $\text{NRV} < 1$ is 0.258 ± 0.044 under the value term and 0.286 ± 0.025 under the control.

The retrieval read-out. Retrieval is computed from the same per-record source: for each parent the 8 displaced geometries are ranked by $\log q_\theta$ and we ask whether the top one, or the top three, contain the geometry GFN2-xTB scores lowest. Chance is $1/8 = 12.5\%$ and $3/8 = 37.5\%$. The read-out is invariant to both the per-group additive constant and any positive rescaling of $\log q_\theta$, so it survives the calibration failure of Section 4.3 intact; it was added after the evaluation was designed and is not a pre-registered endpoint.

Table A7: Retrieval, per seed, on the primary endpoint. Each entry is the fraction of the 93 parents whose lowest-energy displaced geometry is ranked first (top-1) or in the top three (top-3) by $\log q_\theta$. Chance is 12.5% and 37.5%. The teacher row substitutes $-E_{\text{eSEN}}/kT$ for $\log q_\theta$ on the identical geometries and is not a model. Seed numbers are within-arm labels and do not pair across arms; seed 5 of the flow-matching arm is the later-scored control seed of Appendix B.10, and the value arm has four seeds, so its seed-5 cells are empty.

Arm	seed 1	seed 2	seed 3	seed 4	seed 5	mean
Flow matching only, top-1	23.7%	19.4%	20.4%	15.1%	22.6%	20.2%
Value (ours), top-1	26.9%	33.3%	28.0%	30.1%	—	29.6%
Flow matching only, top-3	49.5%	49.5%	50.5%	47.3%	50.5%	49.5%
Value (ours), top-3	64.5%	65.6%	62.4%	57.0%	—	62.4%
<i>Teacher</i> (not a model)	—	—	—	—	—	87.1% / 98.9%

The teacher-evaluator agreement reference, and why it must be re-measured on any other ensemble. The reference is measured, not assumed: the identical geometries are scored with the eSEN teacher (Fu et al., 2025), $-E_{\text{eSEN}}/kT$ is substituted for $\log q_\theta$, and the result is pushed through the same per-group code. One reference run serves every arm of the round, because their geometries are byte-identical. On the 93 primary parents with the data geometry dropped it gives $r = 0.927$; the same construction over all 120 parents gives 0.933, quoted here for context only and never mixed with a model number computed on a different population. The value arm’s 0.369 is therefore about 40% of the teacher-evaluator agreement reference on this ensemble in absolute terms, and closes 23.3% of the gap to it. A model could in principle align with GFN2-xTB better than the teacher does, so this is a reference and not an attainable maximum.

The reference is high because the ensemble is wide, which is a property of the ensemble rather than of the method. On the 93 primary parents with the data geometry dropped the within-group spread of $E_{\text{GFN2-xTB}}$ is 14.3 eV on average (median 10.7 eV, maximum 50.2 eV), while the teacher-evaluator disagreement on *relative* energies inside a group is 4.15 eV mean absolute (median 2.19 eV). Two potentials that disagree by a few eV will still rank a fifteen-eV window consistently while saying much less about agreement inside a thermally accessible one, so a narrower perturbation ensemble has a materially lower reference. We therefore state the rule rather than a projection: every new ensemble needs its teacher-evaluator agreement reference measured directly, and no correlation computed on one ensemble may be read as a fraction of a reference measured on another.

The same calibration diagnostics scored with the training teacher. A calibration failure measured with an independent potential invites one specific objection: that what is being measured is disagreement between the evaluator and the teacher the model was trained against, rather than a property of the learned density. The records answer it directly. The eSEN teacher’s energies for all 1080 held-out geometries are on file, so the calibration diagnostics can be recomputed on the *same* 93 parents, the same displacements and the same model log-densities, with the scorer as the only thing that changes. To keep the two columns produced by one code path, each arm’s per-record file was copied with its energy column replaced by the teacher energy for the matching (parent,

displacement) pair, and the paper’s own calibration analysis was then run on the copies with the reference geometry dropped and $kT = 1.0$ eV, exactly as for the reported column; the 93-parent restriction and the median-over-parents-then-mean-over-seeds aggregation are the same routines that produce Table A5. Run this way the pipeline returns the reported GFN2-xTB column unchanged to the last printed digit — $r = 0.200/0.369$, $\text{NRV} = 1.402/2.144$, $\text{NRV}_{\min} = 0.825/0.747$, slope = $0.130/0.448$ and $kT_{\text{eff}} = 8.02/2.34$ eV — which is what licenses reading the second column beside it. **No GFN2-xTB number in this paper changes.**

Table A8: The calibration diagnostics of Table A5 recomputed with the training teacher in place of the independent evaluator. Same 93 primary parents, same displaced geometries, same model log-densities, same $kT = 1.0$ eV, same aggregation (median over parents within a seed, then mean over seeds $\pm$ the standard error over seeds; r and $\text{NRV}_{\min}$ are means over parents). Only the energy column differs. The GFN2-xTB block is the reported primary endpoint and is reproduced here by the identical code path. kT_{eff} ideally equals the 1.0 eV training target; a constant log-density scores $\text{NRV} = 1$ exactly, so no arm under either scorer clears that bar.

Arm	seeds	$r \uparrow$	$\text{NRV} \downarrow$	$\text{NRV}_{\min} \downarrow$	slope	kT_{eff} (eV)
<i>Scored with GFN2-xTB — the independent evaluator (the reported primary endpoint)</i>						
Flow matching only	5	0.200 ± 0.018	1.402 ± 0.038	0.825 ± 0.009	0.130 ± 0.012	8.02 ± 0.81
Value (ours)	4	0.369 ± 0.023	2.144 ± 0.410	0.747 ± 0.014	0.448 ± 0.051	2.34 ± 0.30
<i>Scored with eSEN — the potential the value term was trained against</i>						
Flow matching only	5	0.202 ± 0.018	1.676 ± 0.025	0.821 ± 0.009	0.189 ± 0.017	5.52 ± 0.67
Value (ours)	4	0.399 ± 0.027	2.821 ± 0.576	0.716 ± 0.018	0.718 ± 0.043	1.41 ± 0.09

Three things follow. First, the raw normalised residual variance worsens against the training teacher as well, and by more than against the independent evaluator: $1.676 \rightarrow 2.821$ under eSEN against $1.402 \rightarrow 2.144$ under GFN2-xTB. The failure to calibrate is therefore not an artefact of transferring from the teacher to a different potential; it is present against the very potential the model was trained against, and the ordering-without-calibration reading of Section 4.3 holds under both scorers. Second, the two endpoints separate more sharply against the teacher, not less: the slope rises from 0.448 to 0.718 and kT_{eff} moves from 2.34 eV to 1.41 eV against the 1.0 eV numerical target, while NRV moves the other way. That is the mechanism Section 4.3 states — the direction of the density’s response and its optimal rescaling improve while the residual that no single rescaling can remove grows — and seeing it more clearly under the teacher corroborates it. Third, the ordering endpoint is close to scorer-independent for the control (0.200 against 0.202) and somewhat higher for the value arm (0.369 against 0.399), so the independent evaluator is if anything the more conservative choice for the comparison this paper reports, and it remains the primary one for the reason given in Appendix B.5: it was not used to train the model.

B.8 PERTURBATION FAMILIES

The reported groups are isotropic Gaussian displacements, which carry a confound that this subsection quantifies and partly removes.

What every reported number uses, and the confound it carries. All reported evaluations use $x^{(k)} = x^* + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ per coordinate and $\sigma = 0.15$ Å, plus the unperturbed data geometry; the training shards have the same character, with $K = 5$ geometries per parent at $\sigma \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å. Because $\|\varepsilon\|$ concentrates only as $O(1/\sqrt{3N})$, the members of a group sit at *different* distances from the data geometry: the mean root-mean-square deviation is $\sigma\sqrt{3} \approx 0.25$ Å and the within-group coefficient of variation of that RMSD is 8.1%. Both $\log q_\theta$ and E fall off with that distance, so a model that had learned nothing beyond “further from the data, lower density” would score a positive per-group correlation carrying no energy information at all. That is why the confound has to be broken by construction rather than adjusted for: the nuisance variable is held exactly constant in the fixed-RMSD family below, so a residual contrast there cannot be a distance effect.

The generators, and which were evaluated. Four perturbation generators are implemented in pure array code — `fixed_rmsd`, `torsion`, `bond_angle` and `normal_mode` — of which only `fixed_rmsd` and `normal_mode` were used in any evaluation reported in this paper, and no result for the other two appears anywhere. `fixed_rmsd` draws a random direction, projects out

the six rigid-body modes, Kabsch-aligns the displaced geometry to the reference and rescales the displacement by scalar bisection until the aligned RMSD equals the requested value; a unit test over a set of molecules finds the realised RMSD reproducing its target with standard deviation 1.3×10^{-16} Å, against the 8.1% coefficient of variation of the Gaussian family it replaces. The test covers that the shell is exactly a shell; it does not establish, and we do not claim, that a shell is a physically preferable ensemble. `normal_mode` displaces along a thermal-like combination of low-frequency modes from an anisotropic network model.

Status of the controlled evaluations. On fixed-RMSD shells, with four seeds per arm, the value arm reaches +0.305 against the flow-matching control’s +0.182, a difference of $\Delta r = +0.124$: the effect survives holding the geometry distance exactly constant, at a reduced magnitude. On normal-mode displacements, with five seeds per arm, the difference is $\Delta r = +0.185$, with a seed spread wide enough that the two arms’ observed ranges are not separated; it is reported as supporting context only. The torsional and bond-angle families, and the partial correlation $r(\log q_\theta, -E \mid \text{RMSD})$ that would quantify the residual radial component inside the Gaussian family itself, have not been run, so the confound is bounded rather than excluded (limitation (v)). One caveat travels with all of these: by the reference argument of Appendix B.7, a narrower perturbation ensemble lowers the measurable teacher-evaluator agreement reference, so each controlled family would need its own directly measured reference before its correlations were read as a fraction of one. The two contrasts above are internally paired comparisons and nothing more.

B.9 THE MATCHED SIX-CELL GRID

The grid is the experiment that isolates what the energy term contributes, among the runs that completed. Six cells at five seeds each are matched in every respect except the physics block, so that the differences between columns are attributable to the physics block and to seed noise and to nothing else in the pipeline.

Configuration. Thirty training runs share: batch size 8 with two accumulation steps, 30 000 optimiser steps, $\lambda_E = 3 \times 10^{-5}$ where active, $\lambda_1 = 0.1$ where active, $\lambda_3 = 0$, parents per value step $M = 8$, value-loss cap 1500, the value term every fourth step on parents of at most 50 atoms, the integrator and Hutchinson settings of Table A1, *no* on-policy distillation in any cell, and perturbation shards built from the *training* split only, so that no cell reads energies for a molecule the evaluation will see. All 30 runs warm-start from one common checkpoint at the end of epoch 0, so seed variation is variation in the physics-carrying phase of training and not in the flow-matching pre-phase. Table A9 gives what differs, together with the complete launch record.

Results, per seed. Table A10 gives every completed evaluation of the grid. Each entry is one training run, scored by the protocol of Appendix B.5 — 120 held-out parents, one data geometry plus 8 Gaussian displacements at $\sigma = 0.15$ Å per parent, the data geometry *dropped* before the per-group Pearson correlation is formed, GFN2-xTB as the independent scorer, and $\log q_\theta$ obtained by an ODE integration of the clamped positional flow with Hutchinson divergence estimation (Appendix A.2) — and the cell entry is the unweighted mean over that cell’s completed seeds with the standard error over seeds.

The six contrasts the grid was built to run. Each is the difference Δr between the seed means of two cells of Table A10, quoted with the completed-seed counts behind it; the question each answers was fixed when the grid was designed and is the reason that cell exists. Per-seed values for every cell are in Table A10 and every one of them is drawn in Figure 2. (1) *Does the value term work at all?* D versus A: $\Delta r = +0.204$ (3 vs 5 seeds) — yes; every completed D seed (0.398–0.452) lies above every completed A seed (0.197–0.234). (2) *Does it need the energy information, or only the energy values?* D versus C, the *y*-randomisation control that keeps each parent’s energy multiset and within-parent spread and destroys only the geometry-to-energy pairing: $\Delta r = +0.236$ (3 vs 5) — it needs the pairing, and again the two observed seed ranges do not overlap (0.125–0.232 for C). (3) *Is it more than a density regulariser?* D versus B, where the shard’s energy column is identically zero so the loss is exactly $\text{Var}_k[\log q_\theta]$: $\Delta r = +0.234$ (3 vs 4) — yes, with B spanning 0.125–0.243, again below every D seed. (4) *What does the regulariser contribute on its own?* B versus A: $\Delta r = -0.030$ (4 vs 5) — **nothing measurable**: the point estimate is negative and the two

Table A9: The matched grid: configuration and launch record. All six cells share every hyperparameter listed in the text above; the table gives only what differs. “Shard” names the perturbation file the value term reads, all three built from the training split of the same 30 000 parents and differing only in the energy column. The three status columns are the launch record of the *matched five-seed design* — every seed launched into it is counted and a diverged seed stays in the denominator — and are themselves a result, discussed below and in Appendix B.15. Fourteen further runs of the same design were launched afterwards and are reported separately below rather than folded into these counts. Reading a cell by its completions alone would condition on completion, which is not independent of the cell; per-cell and per-seed results are Table A10 and the two tables must be read together.

Cell	λ_1	λ_E	Shard energies	What it isolates	seeds		
					launch	compl.	diverged
A fmonly	0	0	<i>no shard read</i>	baseline: flow matching alone	5	5	0
B flat	0	3×10^{-5}	<i>identically zero</i>	loss degenerates to $\text{Var}_k[\log q_\theta]$: density regulariser, no energy label	5	4	1
C scram	0	3×10^{-5}	permuted <i>within</i> parent	y -randomisation: same multiset and spread, pairing destroyed	5	5	0
D energy	0	3×10^{-5}	true teacher	the main effect	5	3	2
E force	0.1	0	<i>no shard read</i>	gradient term, no distillation confound	5	5	0
F both	0.1	3×10^{-5}	true teacher	gradient + value	5	2	3
<i>total, cells without the value term (A, E)</i>					10	10	0
<i>total, cells with the value term (B, C, D, F)</i>					20	14	6

Table A10: The matched grid, per seed and per cell. Mean per-group Pearson r between $\log q_\theta$ and $-E_{\text{GFN2-XTB}}/kT$, 120 parents, **data geometry excluded**. Every cell shares backbone, budget, optimiser, warm start, evaluation protocol and training-split-only perturbation shards; only λ_1 , λ_E and the shard’s energy column differ. The seed counts here are *completions*: the launched/completed/diverged record is Table A9 and must be read with this table, since completion is not independent of the cell (limitation (viii)). The means of cells B, D and F are conditional on completion.

Cell	λ_1	λ_E	Energy signal	per-seed r	mean $\pm$ SEM
A fmonly	0	0	none (no shard)	0.234, 0.200, 0.232, 0.221, 0.197	+0.217 $\pm$ 0.008
B flat	0	3×10^{-5}	all energies zero	0.233, 0.125, 0.147, 0.243	+0.187 $\pm$ 0.030
C scram	0	3×10^{-5}	scrambled within parent	0.232, 0.224, 0.168, 0.125, 0.174	+0.185 $\pm$ 0.020
D energy	0	3×10^{-5}	true teacher	0.452, 0.398, 0.412	+0.421 $\pm$ 0.016
E force	0.1	0	none (no shard)	0.105, 0.130, 0.044, 0.086, 0.058	+0.085 $\pm$ 0.016
F both	0.1	3×10^{-5}	true teacher	0.179, 0.210	+0.194 $\pm$ 0.016

arms’ seed ranges overlap almost completely. (5) *Does the gradient term work with the distillation confound removed?* E versus A: $\Delta r = -0.132$ (5 vs 5) — it is below the physics-free baseline, with every E seed (0.044–0.130) below every A seed. (6) *Does the gradient term help on top of the value term?* F versus D: $\Delta r = -0.227$ (2 vs 3) — it does not; both F seeds lie below both the D range and the A range. Two cautions travel with all six. Every one of them rests on two to five completed seeds per cell, so the separations above are statements about the observed seed ranges of small samples and not asymptotic inferences; the arm means and their standard errors are in Table A10. And contrasts (1)–(3) and (6) involve cells D or F, whose means are over survivors. The three cells that completed at 5/5, 5/5 and 4/5 — A, E and the zero-energy B — are the ones entering contrasts (4) and (5), the two negative or null results, which are therefore the least exposed to selection.

What the grid says, and what it does not. Only the true geometry-to-energy correspondence improves local ordering: the true-energy cell clears all three of its controls by 0.20–0.24 in mean per-group r , while the three controls — no physics at all, a pure density penalty with no energy label in it, and real energies attached to the wrong geometries — sit within 0.032 of one another. The grid does *not* say that the value term puts the density on the Boltzmann scale (it does not; Section 4.3),

that the ordering it buys is close to what the teacher itself would score (in absolute terms it is about 40% of the teacher-evaluator agreement reference; Appendix B.7).

The selection risk the launch record creates. Completion is not independent of the cell, and it is lowest exactly where the energy content is highest (3/5 for D, 2/5 for F). If the propensity to diverge is correlated with the quantity the grid measures — plausible in either direction, since a run that drives $\text{Var}_k[\log q_\theta]$ hardest is both the one most likely to break and, on our own account of the mechanism, the one most likely to move the metric — then any cell mean computed over survivors is biased by an unknown amount and in an unknown direction, and we have no correction for that we trust at five launched runs per cell. What we do instead is procedural, and the three rules were fixed before the grid finished launching. (i) Every launched seed is reported, a diverged seed as a diverged seed rather than as an absence. (ii) No diverged seed is replaced within a cell’s reported five, and no cell mean in Table A10 is computed over a re-rolled sample; seeds launched later are reported as a separate block with their own counts, never merged into the matched design. (iii) No cell mean is presented over its completed seeds alone as though it were the cell’s value; every quoted cell mean carries all three counts and is labelled as conditional on completion. The rules cost us: the honest form of the headline contrast is “three surviving seeds of five against five of five” rather than a bare Δ .

Two observations bear on how much to discount, and they do not point the same way. In D’s favour: its three survivors are 0.452, 0.398 and 0.412, and the lowest of them clears the highest value reached by *any* of the fourteen completed control seeds in cells A, B and C (0.243) by 0.155, so for selection alone to manufacture contrast (1) the two lost seeds would have to have landed below every control seed — a strong assumption about runs that failed on a numerical guard rather than on the metric. Against D: the mechanism that makes a run diverge is, on our own account (Appendix B.15), pressure on $\text{Var}_k[\log q_\theta]$, which is exactly the pressure the effect is attributed to, so “diverged” and “would have scored high” are plausibly positively correlated and the survivors could be the tail rather than the body. We cannot decide between these at three seeds and we do not: the number stands as a completion-conditional mean, and contrasts (4) and (5), which involve no cell that lost more than one seed, are the ones a sceptical reader can take at face value.

The later relaunch block, and what it bounds. After the matched grid was scored, a further nine seeds of cells D and F were launched under byte-identical configurations (D seeds 6–9, F seeds 6–10), together with five seeds of a seventh cell E2, identical to E except that the gradient weight is $\lambda_1 = 0.01$ rather than 0.1. They are excluded from Tables A9 and A10 and from Figure 2, which report the matched five-seed design, and are reported here instead so that the complete launch record is visible. All four D seeds completed and all four have been scored under the identical protocol, giving 0.353, 0.374, 0.452 and 0.471, a later-block mean of $+0.412 \pm 0.029$; across all seven scored D runs the mean is $+0.416 \pm 0.017$ and every seed remains above every seed of A, B and C. Four of the five F seeds diverged; the fifth completed and scores 0.201, so all three scored F runs give $+0.197 \pm 0.009$. All five E2 seeds completed and give 0.118, 0.076, 0.089, 0.078 and 0.038, a mean of $+0.080 \pm 0.013$.

Three things follow. Over the whole campaign cell D is 9 launched, 7 completed, 2 diverged and cell F is 10 launched, 3 completed, 7 diverged, so the completion rates reported above understate how badly the combined cell fails to train. Every scored D seed, relaunched or not, lies above *every* one of the fourteen completed control seeds in cells A, B and C, whose maximum is 0.243, so the worry that D’s survivors are a lucky subset is bounded by data rather than by argument. And the E2 cell shows that the gradient term’s negative result is not an artefact of one weight — a tenfold smaller λ_1 still lands at $+0.080$, below cell A’s $+0.217$ — while adding five more value-free runs with no divergences to the stability record.

B.10 PROVENANCE AND AUDIT OF THE EARLIER CHECKPOINT FAMILY

The checkpoints that produce Table A4 and Table A5 come from an earlier round whose *contrasts* carry four defects: an on-policy distillation term present only in the gradient-supervision arms, a perturbation shard drawn from the split the evaluation parents come from (item (a) of Appendix B.4), a negative control whose validation shard was never scrambled (item (b)), and cells that differ in batch size, parents per value step or loss cap as well as in the physics block. Which comparisons survive

must be stated separately rather than in one verdict. The value-versus-flow-matching direction and approximate size survive as a *corrected held-out evaluation of this checkpoint family*, which is what Tables A4 and A5 report: the shard overlap is removed by restricting to the 93 disjoint parents and the high-leverage geometry by dropping it, and both corrections reduce the contrast. The round’s four-cell contrasts in Table A11 do not survive and are superseded by the matched grid; its force comparisons are confounded by the distillation term and are superseded by cell E, its scrambled control by cell C. We keep the record because the provenance of the reported checkpoints matters and because a defect record that disappears once the defect is repaired is not a record.

Table A11: The earlier checkpoint family, and the flags that bound which of its contrasts remain usable. Metric is the mean per-parent Pearson correlation between $\log q_\theta$ and $-E/kT$ from GFN2-xTB, with the data geometry *included* and 30 parents per cell — a different population and a different protocol from every other table in this paper. **No number in this table may be compared with any number outside it:** the only admissible comparator for each cell is the same round’s own control, named in the last column. Per-seed values recomputed for this table from the round’s own per-record evaluation outputs. The four-cell *contrasts* of this table are superseded by Table A10; the round’s checkpoints are not, and are re-evaluated on the primary endpoint in Tables A4 and A5.

Cell	Physics	parents	seeds	per-seed r	admissible comparator
a1	none (flow matching only)	30	2	0.223, 0.126	—
a2	gradient supervision only	30	1	0.109	a1, this round only
a3	value supervision only	30	2	0.420, 0.424	a1, this round only
a4	gradient + value	30	1	0.294	a3, this round only
<i>Confound flags</i>					
a2, a4	carry an on-policy distillation term (weight 0.1, forces queried at model samples every eighth step) in addition to the off-policy endpoint term, so “gradient supervision” there is force matching <i>plus</i> distillation.				
a4	uses a single parent per value step and no value-loss cap — the configuration diagnosed as pathological in Appendix B.15.				
a3	trains on a perturbation shard built from the validation split the evaluation parents are drawn from; a1 reads no shard (item (a) of Appendix B.4).				

The arms that supply the reported checkpoints are the same round’s value and flow-matching cells re-evaluated on the primary endpoint, with per-seed values in Table A5. Their estimator configuration is not uniform: all four value seeds and two of the three scrambled seeds run $M = 4$ with cap 3000, and the third scrambled seed runs $M = 8$ with cap 1500 and batch 4, which is the one reported cell that is not configuration-matched to its comparator. That seed is additionally a *replacement*: two seeds of the scrambled arm diverged before their first checkpoint and were relaunched under the higher- M configuration ($M = 8$, cap 1500), which is a change of estimator settings and not a demonstrated stabilisation (Appendix B.15), so the assurance that no diverged seed is replaced holds of the matched grid and not of this arm.

Two recovered evaluations, and what each does to the reported numbers. Two completed checkpoints of this round carried no primary-endpoint score when the tables were first assembled — one because its evaluation job was submitted against the wrong output directory and died, one because it was simply never submitted — and neither absence had been recorded. Both have since been scored, and both absences had flattered the reported contrast, so the correction runs against us in one case and in our favour in the other.

A fourth scrambled seed. The recovery uses no new computation: the same checkpoint had been scored under a wider 240-parent protocol whose parents and displacements are byte-identical to the primary round’s for the first 120 group identifiers (agreement in $\log q_\theta$ to 1.2×10^{-4}), so restricting those records to the 93 disjoint parents and pushing them through the same calibration analysis recovers the missing value; the route reproduces the two already-published seeds of this arm to six decimal places, which is what licenses it. The recovered seed scores $r = 0.161$ with slope 0.104 and $kT_{\text{eff}} = 9.59$ eV — the lowest of the four — so the arm at four seeds sits at 0.210 rather than the 0.226 over three seeds printed in Tables A4 and A5. We leave the three-seed value in the tables, since that is the set scored end-to-end under the primary protocol, and record the fourth here with its provenance; either way the arm is the defective control of Appendix B.4 and is superseded by cell C.

A *fifth flow-matching seed*. All five control-arm seeds reached the full 30 000-step budget with finite weights; only four had been submitted to the primary round’s evaluation. The fifth has since been scored under the identical protocol — the same 120 parents and displacements, the same evaluation seed, the same density estimator — so what changed is an evaluation of a checkpoint that already existed, not a new training run. The earlier absence was a defect rather than a wash: it was the control arm’s only unreported run, whereas the value arm’s shortfall was a divergence, and under the superseded 30-parent round of Table A11 this same checkpoint had scored the higher of that round’s two flow-matching values, so its absence could only have flattered the contrast. On the primary endpoint it scores $r = 0.192$ with $\text{NRV} = 1.388$, slope 0.139 and $kT_{\text{eff}} = 7.18$ eV, in the middle of its arm. Including it moves the control mean from 0.202 to 0.200 and tightens the standard error over seeds from 0.023 to 0.018; the difference between the arms therefore grows from $\Delta r = +0.167$ to $\Delta r = +0.169$, and with the fifth seed included all four value seeds lie above all five control seeds. Every flow-matching entry in Tables A4 to A7 and in Figures 3 and 4 is computed over all five seeds. The recomputation was validated by first reproducing every stored four-seed value to three decimals — the per-seed correlations, NRV, NRV_{min} , the slope, the scale ratio, kT_{eff} , the 69-of-93 per-parent count and both retrieval rates — and only then adding the fifth seed, which moves that per-parent count to 67 of 93.

B.11 A LARGER-POPULATION REPLICATION, AND THE CONTAMINATION IT CARRIES

The same round’s checkpoints were also scored on a wider 240-parent draw from the same held-out split, whose parents and displacements agree with the primary round’s on the first 120 group identifiers to 1.2×10^{-4} in $\log q_\theta$. Mean per-group r , data geometry dropped: the flow-matching control reaches $+0.2220 \pm 0.0082$ over four seeds and the value arm $+0.4172 \pm 0.0088$ over three, a difference of $\Delta r = +0.1952$; the arm trained on scrambled energies reaches $+0.2284 \pm 0.0102$ over two seeds and its two relaunched seeds $+0.2446 \pm 0.0500$, in both cases essentially where the control already sits. Direction and mechanism reproduce on the larger population.

This family is nonetheless recorded here **together with, and inseparable from, the reason it cannot be the headline**. The 93-parent restriction is not a convenience that a larger population improves upon; it is a contamination control. One of the two perturbation shards read by every value-supervised arm of this round was built from the split the evaluation parents are drawn from, while the control reads no shard at all (item (a) of Appendix B.4), so the overlap is arm-specific and favours the value arm, and a 240-parent population contains *more* of it. A larger sample of a contaminated population is a more precise estimate of a contaminated quantity, not a more trustworthy one, and that the contrast here ($+0.1952$) exceeds the primary endpoint’s ($+0.169$) is what the confound predicts. The headline remains the 93-parent primary endpoint, $\Delta r = +0.169$ over four value and five control seeds (Table A4). Nothing in this subsection is quoted in the abstract or the main text, and no claim the paper makes rests on it.

B.12 COMPLETE RECORD OF EVALUATIONS RUN

This subsection exists so that a reader can check that nothing that completed is hidden. Every completed Boltzmann evaluation held in the run store falls into exactly one of the families listed below, each with what it is, how many completed evaluations it holds, the parent population it was scored on, its value, and its disposition: reported in this paper, superseded by a named later family, or deliberately set aside for a stated reason.

Two conventions, and why the two halves of the record cannot be compared. Families scored under the current protocol report the per-group Pearson r with the high-leverage data geometry *dropped* (the paper’s convention, Appendix B.5). The older 30-parent rounds have only the statistic with the data geometry *included*, on a different and much smaller parent population. Numbers marked † below are of that second kind: they are recorded for completeness and **may not be compared with any other number in this paper**, in either direction.

Cells D and F contain later-block evaluations that lie outside the matched five-seed design. Figure 2 and Table A10 retain only the prespecified matched-block means, $+0.421 \pm 0.016$ for D (5 launched / 3 completed / 2 diverged) and $+0.194 \pm 0.016$ for F (5 / 2 / 3). The later-block and campaign-wide aggregates are reported separately in Appendix B.9: $+0.416 \pm 0.017$ over all seven scored D runs (9

Table A12: Families whose evaluations are used somewhere in this paper. "Evals" counts completed evaluations present in the run store, not launched runs; the launched/completed/ diverged records are Table A9 and Section 4.2. The value column aggregates over *all* completed evaluations of the family, which is a different aggregation from the matched-design means the paper reports: for cells D and F the design means over the matched five-seed block are +0.421 and +0.194 (Table A10), and those, not this column, are what the paper's contrasts use. No contrast is formed from this column. † marks the data-geometry-included convention on 30 parents (see text).

Family	What it is	Evals	Parents	Value (r)	Disposition
clean.p0.A.fmonly	matched grid, cell A: flow matching only	5	120	$+0.2167 \pm 0.0078$	reported, Table A10
clean.p0.B.flat	cell B: shard energies identically zero	4	120	$+0.1871 \pm 0.0298$	reported, Table A10
clean.p0.C.scram	cell C: energies scrambled within parent	5	120	$+0.1847 \pm 0.0196$	reported, Table A10
clean.p0.D.energy	cell D: true teacher energies	7	120	$+0.4160 \pm 0.0166$	reported, Table A10 and the later block
clean.p0.E.force	cell E: gradient term, $\lambda_1 = 0.1$	5	120	$+0.0845 \pm 0.0156$	reported, Table A10
clean.p0.F.both	cell F: gradient + value	3	120	$+0.1967 \pm 0.0094$	reported, Table A10 and the later block
clean.p0.E2.force	cell E2: gradient term, $\lambda_1 = 0.01$	5	120	$+0.0799 \pm 0.0127$	reported, Section 4.1
wide.a1.fm.only	primary-endpoint control arm	5	120 (93)	$+0.200 \pm 0.018$ on 93; $+0.2154$ on 120	reported, Table A4
wide.a3.energy.only	primary-endpoint value arm	4	120 (93)	$+0.369 \pm 0.023$ on 93; $+0.3968$ on 120	reported, Table A4
wide.a6.energy-only-shuffled(.stab)	scrambled control of that round	3	120	$+0.2331$ (2), $+0.2957$ (1)	reported, Section 4.2; defective, superseded by cell C
abl.a1..a6	the earlier 30-parent round	9	30	† $+0.109$ to $+0.422$	Table A11; contrasts superseded by the matched grid
n240.a1/a3/a6(.stab)	240-parent replication	4/3/ 2/2	240	$+0.2220$ / $+0.4172$ / $+0.2284$ / $+0.2446$	Appendix B.11: supporting only, confounded

launched / 7 completed / 2 diverged over the whole campaign) and $+0.197 \pm 0.009$ over all three scored F runs (10 / 3 / 7). No matched contrast is formed from the campaign-wide value column of Table A12.

The set-aside families' numbers, recorded so that they are not absent. The three 30-parent families below are in the data-geometry-included convention (†) and none may be compared with any other number in this paper; the re-scorings under alternative integrator settings are values of checkpoints already reported in Table A12, and no value of theirs is quoted anywhere in this paper. Transition metals: $+0.365$ and $+0.329$ for the value arm and -0.158 for the force-only arm. Configuration generations preceding the matched grid, where a value is on record: $+0.413$ (v3), $+0.400$ and $+0.322$ (v3b), $+0.091$ (v4), $+0.172$ (v4-stable) and -0.082 (on-policy at epoch 0). The capped-step sweep: -0.396 , -0.326 , -0.221 , -0.051 and -0.104 , all five negative, for the reason given in Table A13.

B.13 DE NOVO GENERATION QUALITY

An ordering result is only useful if the samples are still molecules. Table A14 reports bond-free de novo generation quality on OMol25 for the value-supervised arm and for our own bond-free flow-matching control trained under an identical budget: the fraction of samples for which xyz2mol returns a sanitisable molecule, the fraction that are a single connected component, the fraction

Table A13: Families that completed and are deliberately not used for any claim. Each is named, counted and given its reason, so that no completed evaluation is silently absent. † marks the data-geometry-included convention on 30 parents, which is not comparable with any other number in this paper.

Family	What it is	Evals	Parents	Why it is not used
OOD.tmqm.*	transition-metal out-of-distribution probes	3	30	spin multiplicity is not passed to the teacher (Assumption 6); consistently with Section 2, no transition-metal result is reported
v2.forceonly, v3.energy, v3b.energy_hi (epoch0), v4, v4_stable (tmqm), onpolicy_4m.epoch0 eval_sweep/*	configuration generations preceding the matched grid	8	30	superseded by the matched grid: different training configurations, not comparable to it or to one another
	the first capped-step sweep	5	30	run before the step-based λ schedule was fixed, so every arm was effectively flow matching only — which is why all five are negative
eval_res/*	re-scorings, under alternative integrator settings, of checkpoints already listed in Table A12	49	120	no training run and no checkpoint of its own; the analysis these supported is not part of this paper and no claim here is formed from them

passing the PoseBusters geometry checks (Buttenschoen et al., 2024), and uniqueness. Both rows are recomputed for this table from the per-run evaluation records; samples on which the bond-perception step times out are counted as failures.

The comparison is deliberately narrow, and we state its scope rather than leaving it to be inferred: one method and one control, trained on the same corpus under the same budget and scored by one script. We tabulate no published generator’s numbers, because they are produced on different corpora, with bond supervision, and under different sampling and scoring protocols, and placing them in the same table would invite exactly the comparison this paper does not make. Three caveats attach to the table and no entry in it is bolded. First, at $n = 100$ samples per model the binomial standard error on each percentage is about 5 points, so differences of a few points are not resolved: the value arm’s 43.0 against the control’s better seed at 41.0 is noise, and the control’s own two seeds are 41.0 and 30.0. Second, the value row is a single run while the control row is a two-seed mean, so this comparison is not seed-controlled. Third, generation quality is not the axis on which the arms differ: their local density–energy ordering differs by $\Delta = +0.169$ on the primary endpoint, while their validity rates differ by less than the control’s seed-to-seed spread. What the table supports is the absence of evidence that value supervision degrades generation quality, and nothing more.

Table A14: Bond-free de novo generation quality on OMol25, in per cent, $n = 100$ samples per model scored with one script. Validity is bond perception plus sanitisation; connectivity is the single-component fraction; PoseBusters is the geometry-check pass rate. At $n = 100$ the binomial standard error is about 5 points per cell, so nothing here is bolded and differences of a few points should not be read as differences. The value row is the arm whose ordering is reported in Table A4; the control row is the mean of its two evaluated seeds, whose individual values are given beneath it.

Model	valid	connected	PoseBusters	unique
Value supervision (ours)	43.0	31.0	20.0	100.0
Bond-free flow-matching control (ours, 2 seeds)	35.5	23.0	16.0	100.0
seed 1	41.0	27.0	17.0	100.0
seed 2	30.0	19.0	15.0	100.0

B.14 THE CAPABILITY MATRIX

Table A15 places the configuration studied here among the model classes it borrows from. The point is not that it scores higher, but that most entries are quantities the other classes do not define, which is

why the measurement of Section 4.2 has no direct precedent to be compared against. The likelihood column classifies the *kind* of density each class defines rather than asserting exactness for any of them, ours included: what we compute is a finite-step discretisation of a continuous-flow likelihood with a stochastic trace, whose accuracy — not whose identity — is fixed by that discretisation (Appendix A.2). 3D molecular generators (Hoogetboom et al., 2022; Xu et al., 2023; Dunn & Koes, 2025; Morehead et al., 2026; Daigavane et al., 2024) assemble composition and geometry de novo but are not trained against a transferable energy and report no density–energy check, while conformer models and Boltzmann samplers (Jing et al., 2022; Klein & Noé, 2025; Tan et al., 2025; Rehman et al., 2026; Havens et al., 2025) have tractable densities — and, for Adjoint Sampling, a universal-potential energy target — but receive the molecular graph and therefore do not generate composition.

Table A15: Capability matrix. Entries record what each model class *can be asked*, not how well it scores; “n/a” marks a question that is undefined for that class, “n.r.” a capability the authors do not report, “graph” a model that is given the molecular graph, and “FF” a classical force field as the energy target. The likelihood column records the kind of density a class defines: “bound” a variational bound, “CNF” a continuous-flow likelihood (exact in continuous time; what any implementation reports is a discretisation, ours included). Methods without a tractable likelihood cannot be given a density–energy measurement at all, so the last column states which question each class can be asked rather than which model is better. The Elements column records the element coverage of each model’s training corpus, not a demonstrated evaluation scope: every result in this paper is evaluated on main-group organic parents at the singlet default (Assumption 6).

	Ele- ments	Bond labels	De-novo comp.	Likeli- hood	Energy- trained	Amor- tised	log p vs. E reported
EDM / GeoLDM	~5	implicit	yes	bound	no	yes	no
FlowMol3 (+ our control)	~10	yes (no)	yes	CNF	no	yes	no
Zatom-1	OMol25	no	yes	n.r.	no	yes	no
Torsional diffusion	organic	graph	no	CNF	no	yes	yes
Transferable BG	peptides	n/a	no	CNF	FF	partly	yes
SBG / ArBG	peptides	n/a	no	CNF	FF	partly	yes
Adjoint Sampling	organic	graph	no	CNF	yes	yes	yes
This paper	83 (corpus)	no	yes	CNF	yes	yes	yes

B.15 NUMERICAL STABILITY OF THE VALUE OBJECTIVE

The value objective is numerically fragile. This subsection is the failure record: what failed, what the evidence says about why, and what the configuration changes did and did not buy.

Diagnosis. The within-parent variance estimator averages over M parents per step and its variance falls like $1/M$. With $M = 1$ the loss was observed to jump from 48 to 974 between consecutive steps, producing gradient spikes that drove weights to non-finite values in 3 of 9 seeds. The guard in place at the time tested the *loss* for finiteness, substituted zero and continued training, so a run whose weights had already been destroyed kept stepping the optimiser, kept logging plausible losses and kept writing checkpoints that a later evaluation script would take for healthy; one evaluation was run against a corrupted checkpoint before the pattern was recognised. A guard that masks a non-finite loss and trains on converts a loud failure into a silent one. No arm reported in this paper was trained at $M = 1$; that configuration is the diagnosis, not a reported result.

The three changes, and what they bought. (i) Parents per value step were raised from 1 to 8, lowering the estimator variance by the same factor; the earlier round’s arms sit at the intermediate value 4. (ii) The value-loss cap rejects a step’s value loss outright when it exceeds the cap (1500 in the grid, 3000 in the earlier round) rather than clipping it, since such values are finite and therefore invisible to a non-finiteness test. (iii) A callback sweeps the parameters every 200 optimiser steps and raises immediately if any weight is non-finite, naming the first offending tensors.

The rates are these. In the diagnosis configuration ($M = 1$, no cap), 3 of 9 seeds diverged. In the configuration all thirty grid runs use ($M = 8$, cap 1500), 6 of the 20 launched runs that carry the value term diverged, and none of the 10 that do not. *We therefore claim no mitigation*: an order-of-magnitude reduction in estimator variance plus an outlier rejector leave the failure rate where it was, and we have no configuration of this objective that we can claim is safe. What is demonstrably fixed is the failure’s *visibility*: the callback aborts at the first non-finite parameter and leaves the

last good checkpoint intact, and why every evaluation launcher refuses any checkpoint containing a non-finite parameter.

Where the instability lives. The grid discriminates in both directions, and the attribution it supports is the reverse-time log-density objective and its gradient, rather than the teacher’s energy labels. Divergence *requires* $\lambda_E > 0$: none of the ten runs with $\lambda_E = 0$ diverged, and five of those ten carry the gradient term at $\lambda_1 = 0.1$, so neither a physics term *per se* nor the score read-out of (3) produces the instability. Divergence does *not* require energy labels: cell B reads a shard whose energy column is identically zero, so its loss is exactly $\frac{1}{M} \sum_m \text{Var}_k[\log q_\theta]$, and its lost seed nevertheless drove weights to non-finite values at optimiser step $\approx 25\,800$, late in a 30 000-step run. What survives both cuts is the reverse-time log-density of (1), the second derivatives its divergence term back-propagates, and the within-group variance formed from them. The grid does not separate the estimator from its gradient, and we do not claim it does. The per-cell dose–response is suggestive rather than conclusive at these counts: 1/5, 0/5, 2/5 and 3/5 for B, C, D and F.

Consequence for every reported mean. Because only cells carrying the value term have ever diverged, and their completion rates fall as their energy content rises (5/5 scrambled, 4/5 zero-energy, 3/5 true energies, 2/5 both terms), an arm mean over completed seeds is a mean conditional on completion, and the conditioning is most likely to be informative — the resulting bias largest, in an unknown direction — precisely for the cells this paper’s thesis leans on. That is what the reporting rules of Appendix B.9 exist to keep visible. The objective also costs memory, because the density term differentiates a reverse-time solve.

B.16 INFERENCE STATISTICS, COLLECTED FOR COMPLETENESS

The paper reports effect sizes, arm means with standard errors over independently trained seeds, and every seed-level result, for the reason given in Appendix B.5: several cells contain only two to five completed seeds and completion is not independent of the objective (Appendix B.15), so asymptotic null-hypothesis tests are not the primary evidence for anything claimed here. Inferential statistics were nonetheless computed during the analysis, and Table A16 records *all* of them in one place so that none is selectively present or selectively absent. It is a completeness record: no statement in this paper is made in terms of these quantities, and they are not repeated anywhere else in the manuscript.

Test conventions. Rows marked *Welch* are two-sided Welch unequal-variance t statistics computed on per-seed arm means, so the unit of replication is one training run and the sample size of a cell is its number of *completed* seeds; the Welch–Satterthwaite degrees of freedom lie between 2 and 8 throughout, which is why these statistics are large mostly when the within-cell seed spread is small. Rows marked *permutation* are exact two-sided permutation tests that enumerate every relabelling of the pooled seeds between the two arms, so their p -values are distribution-free but are bounded below by the reciprocal of the number of relabellings (1/126 for four against five seeds). Rows marked *Wilcoxon* are two-sided Wilcoxon signed-rank tests on paired per-parent, seed-median values over the 93 primary parents; their unit is a parent within a fixed set of nine trained models, so they describe how a shift is distributed over the evaluation set and are *not* additional training replications. No p -value here is corrected for multiplicity, and every row involving cell D, cell F or the value arm of the earlier round is conditional on completion.

B.17 LIMITATIONS, IN FULL

(i) *The density is not calibrated, and on the primary endpoint it is worse than the control’s.* Every arm, including the value arm, has median NRV ≥ 1 — 2.144 against the control’s 1.402 — where a constant log-density scores exactly 1, at an implied temperature above 2 eV against the 1.0 eV target. Nor is this merely the wrong evaluation temperature: by (4) the best any single rescaling could reach is $1 - r^2$, close to 1 at these correlations, so the calibration cannot be repaired without first raising the correlation substantially. The learned density has the right direction of response and a dynamic range that overshoots. Scoring the identical geometries with the training teacher instead of the independent evaluator makes raw NRV worse rather than better (1.676 $\rightarrow$ 2.821; Table A8), so the failure is not an artefact of the change of potential. Two diagnostics move the other way and belong beside the headline: the optimally rescaled bound $\text{NRV}_{\min}$ falls from 0.825 to 0.747 and the

Table A16: Every inferential statistic computed for this paper, in one place. Conventions are defined in the text immediately above: *Welch* is a two-sided unequal-variance *t* on per-seed arm means, *perm.* an exact permutation test over seed relabellings, *Wilcoxon* a two-sided signed-rank test on paired per-parent values (93 parents). Δ is the difference of arm means (or, for the Wilcoxon rows, the median paired shift). Seed counts are completed seeds. This table is a completeness record and nothing in the paper is stated in terms of it; the corresponding descriptive results are in Tables A4, A5 and A10. No *p*-value is corrected for multiplicity, degrees of freedom are small throughout, and every row involving a value-carrying cell is conditional on completion (Appendix B.15).

Comparison	n vs n	Δ	test	statistic
<i>Primary endpoint, 93 disjoint parents, data geometry dropped (Table A4)</i>				
value vs flow matching	4 vs 5	+0.169	Welch	$t = 5.82$
value vs flow matching	4 vs 5	+0.169	perm.	$p = 1/126 = 0.008$
adverse imputation (i), at the control mean	5 vs 5	+0.135	Welch	$t = 3.21$
adverse imputation (i), at the control mean	5 vs 5	+0.135	perm.	$p = 0.024$
adverse imputation (ii), at the worst value in any arm	5 vs 5	+0.124	Welch	$t = 2.38$
adverse imputation (ii), at the worst value in any arm	5 vs 5	+0.124	perm.	$p = 0.056$
<i>Matched six-cell grid, 120 parents (Table A10)</i>				
(1) D true energies – A flow matching	3 vs 5	+0.204	Welch	$t = +11.36$
(2) D true energies – C scrambled	3 vs 5	+0.236	Welch	$t = +9.28$
(3) D true energies – B zero energies	3 vs 4	+0.234	Welch	$t = +6.89$
(4) B zero energies – A flow matching	4 vs 5	−0.030	Welch	$t = −0.96$
(5) E gradient only – A flow matching	5 vs 5	−0.132	Welch	$t = −7.60$
(6) F gradient + value – D true energies	2 vs 3	−0.227	Welch	$t = −10.05$
<i>Perturbation families (Appendix B.8)</i>				
fixed-RMSD shells, value vs flow matching	4 vs 4	+0.124	Welch	$t = 4.94$
normal-mode family, value vs flow matching	5 vs 5	+0.185	Welch	$t = 1.58$
<i>Larger-population replication, 240 contaminated parents (Appendix B.11)</i>				
value vs flow matching	3 vs 4	+0.1952	Welch	$t = 16.18$
<i>Per-parent paired diagnostics, 93 parents, value minus control (Appendix B.7)</i>				
per-group correlation r	93	+0.182	Wilcoxon	$p = 1.1 \times 10^{-8}$
regression slope	93	+0.346	Wilcoxon	$p = 2.9 \times 10^{-13}$
scale ratio	93	+0.500	Wilcoxon	$p = 3.2 \times 10^{-16}$
temperature-optimal bound $1 - r^2$	93	−0.055	Wilcoxon	$p = 1.6 \times 10^{-4}$
NRV (higher is worse)	93	+0.476	Wilcoxon	$p = 5.0 \times 10^{-7}$

implied temperature from 8.02 eV to 2.34 eV. An anchor term would constrain the additive group offset, but NRV is invariant to that offset, so no offset-based term can move this limitation; the scale-sensitive objective it would need is not implemented.

(ii) *The value arm reaches about 40% of the teacher-evaluator agreement reference in absolute terms, and closes 23.3% of the gap to it.* The teacher’s own energies score 0.927 on the primary endpoint. Read that as headroom rather than as a score: a model could in principle align with GFN2-xTB better than the teacher does, so it is a reference and not an attainable maximum. What the residual is we cannot say; under-training (item vii), the objective’s own limits and the estimator (item vi) are all live, and the one thing it is *not* is teacher–evaluator disagreement. The reference is ensemble-dependent and must be re-measured, not inherited, on any narrower ensemble (Appendix B.7).

(iii) *Local only: cross-basin relative probability is not reported.* Objective and metric both operate on one perturbation cloud per composition ($\sigma \leq 0.40$ Å in training, 0.15 Å at evaluation). By Theorem 2 and corollary 2 that design has an overlap graph with no edges, so a model can reach zero loss and a per-group correlation of 1 while assigning arbitrary relative mass to distinct conformational basins. We therefore claim local density–energy ordering and not a conditional Boltzmann density. The discriminating experiment — a multi-basin reference ensemble per composition, compared on mode occupancies rather than within-cloud ordering — is not in this paper, and Theorem 2 says what design would make it decisive; the mass-covering correction of Midgley et al. (2023) is the machinery it would need. No claim in this paper rests on cross-basin behaviour in either direction.

(iv) *The correlation certifies shape, not temperature.* Pearson r is invariant under positive affine maps of either variable (Remark 13), and $kT = 1.0$ eV is a numerical scale and not a physical temperature. The calibration endpoint is computed on the same 93 parents as the ordering endpoint, so the two describe one population; neither speaks to a physical temperature, and the retrieval read-out is a secondary within-cloud ranking statistic, not a conformer-search benchmark.

(v) *The radial-distance confound is bounded, not excluded.* Energy correlates with distance from the data geometry, which varies within a group (8.1% coefficient of variation on an RMSD of ≈ 0.25 Å), so a model that had learned only “further from the data, lower density” would score a positive r_m carrying no energy information. Two of the four implemented confound-breaking families are measured: the effect survives at reduced magnitude on fixed-RMSD shells ($\Delta r = +0.124$, four seeds per arm), and the normal-mode family gives $\Delta r = +0.185$ over five seeds per arm, with overlapping seed ranges, and is supporting context only. The torsional and bond-angle families, the partial correlation given RMSD, and a noise-scale sweep have not been run (Appendix B.8).

(vi) *The likelihood estimator has not been validated.* (vi-a) *What is reported is a finite-step estimate of $\log q_\theta$ at $t = 1$, and it is not certified as converged.* Every reported $\log q_\theta$ comes from an ODE integration of the clamped positional flow with Hutchinson divergence estimation (Appendix A.2). That estimate is biased by the explicit-Euler state integration and noisy from the stochastic trace, and we do not assert that it is converged. (vi-b) *The training objective and the evaluation metric are formed by the same estimator, but the value term is computed inside the training loop under a training-time budget.* We do not characterise the estimate the objective works from against the one the metric computes, and we cannot say whether any mismatch attenuates the reported effect, inflates it, or is immaterial. (vi-c) *Neither the estimator’s noise nor its bias has been measured on the reported population.* A repeat-to-repeat variance study on that population, and agreement with an exact divergence, are not reported here; under classical attenuation the reported r would be a lower bound (Remark 14), but that is an argument, not a measurement. The missing measurements are the ordinary numerical ones: a higher-order or adaptive integrator for the state, and a validation of the estimate against an exact reference on real molecules. We have not run them, and no result in this paper reflects them. (vi-d) *The reported density is a position-only flow density.* Equation (1) is integrated with the discrete channels clamped at their data labels for the whole reverse trajectory (Assumption 2). That object is normalised and is what every theorem and every number in this paper refers to, but it coincides with the generator’s own joint conditional density $p_\theta(x | c)$ only if the discrete trajectory is determined by its endpoint; the generator evolves types and charges through a continuous-time Markov chain, so in general the two differ, and we have not quantified the difference.

(vii) *Training budget.* Every arm trains 30 000 optimiser steps at an effective batch of 16, approximately 0.12 data epochs; we do not establish convergence, and under-training is a live alternative explanation for the distance to the teacher-evaluator agreement reference rather than an excuse for it. The value term saw only parents of at most 50 atoms while evaluation parents run to 200; that extrapolation is untested.

(viii) *Confounds, defects and fragility.* The gradient-supervision result is unconfounded — cell E of the matched grid carries no distillation term, five completed seeds of five launched, and 120 evaluation parents — but it remains scoped: what is measured to fail is naive off-policy endpoint force matching read through the flow-matching score identity (3) on a corpus that is not a single-temperature equilibrium sample, and no claim is made about force supervision in general (Guo et al., 2025; Hu, 2026). The literature in which gradient information does work differs from our configuration in exactly the two ways Propositions 2 and 3 name, which is why those propositions are candidate mechanisms rather than isolated causes. The earlier round’s four-cell contrasts remain confounded and are superseded by the matched grid; its checkpoints supply Table A4 under the corrected primary endpoint, which is a different claim (Appendix B.10). Four further defects are recorded rather than repaired, and the first two are repaired only in the matched grid. (a) One of the two shards read by that round’s value arms came from the validation split the evaluation parents are drawn from; the primary endpoint excludes the affected parents, but the shards should be rebuilt and both arms re-run. (b) That round’s scrambled arm’s validation shard was never scrambled and its third seed is not configuration-matched, so that control is partly a positive control; cells B and C of the matched grid replace it. (c) Spin multiplicity is not passed to the teacher (Assumption 6), so open-shell species would be scored in the wrong electronic state; no open-shell result is reported, and any extension to radicals or transition-metal complexes requires a spin-aware pipeline first. (d) External baseline coverage is narrow by construction: conformer metrics of the COV/AMR family

score a strictly conditional problem this model does not solve, so we rank against none of them, and the generation comparison of Appendix B.13 is scored at $n = 100$ samples per arm, where the binomial standard error is of the same order as the differences, so it supports only the absence of evidence that the value term *degrades* generation quality. Finally the objective is numerically fragile: 6 of the 20 launched runs carrying it diverged against 0 of the 10 without it, the configuration changes do not measurably lower that rate, and the zero-energy cell diverged too, which implicates the reverse-time log-density objective and its gradient, rather than the teacher’s energy labels (Appendix B.15). Because divergence is not independent of the physics block, every mean for a value-carrying cell is conditional on completion, and we report launched, completed and diverged counts for every such cell. The implemented objectives are moreover clipped and Huberised variants of the stated ones, the $\mathcal{L}_{\text{value}}$ gradient is the truncated-adjoint surrogate of Remark 1, and rotations are not quotiented (Assumption 3), which the within-parent design mitigates but does not remove.

(ix) *What the compressed main-text statement leaves here.* The conclusion states three bounds — the finite-step clamped positional-flow object, the conditioning of every value-cell mean on completion, and locality together with the absence of raw calibration — and this subsection is where each is developed. Four further qualifications travelled with them and are recorded here rather than in the main text. (ix-a) *The reported gradient is a surrogate.* The gradient of $\mathcal{L}_{\text{value}}$ is the truncated-adjoint surrogate of Remark 1 and not an exact likelihood gradient (Section 2.4), so the optimiser follows a related but distinct descent direction from the one the stated objective defines. (ix-b) *The divergence tally behind “conditional on completion”.* In the matched grid, 6 of the 20 runs carrying the value term reached non-finite weights against 0 of the 10 without it, the zero-energy cell included; that pattern implicates the reverse-time log-density objective and its gradient rather than the teacher’s energies, and the per-cell launched, completed and diverged records are in Appendices B.9 and B.15. (ix-c) *Under-training is a live alternative explanation of every absolute level.* At 30 000 optimiser steps, approximately 0.12 data epochs, no arm is trained to convergence, so the distance of any arm from the teacher-evaluator agreement reference is not attributable to the objective (item vii). (ix-d) *The three extensions the bounds imply.* A scale-sensitive term constraining the magnitude of the response would address item (i); groups tiled along a path, so that the overlap graph acquires edges, would address item (iii) through Theorem 2; and a global mass-covering objective of the kind developed for fixed systems (Midgley et al., 2023) would address the allocation target directly. None is run here.

B.18 MAIN-TEXT FIGURES: PANEL CONSTRUCTION

Figure 1 (schematic). No measured quantity appears in the figure. (a) A bond-free de novo flow samples a composition c and a geometry x jointly; with the endpoint discrete labels clamped, reversing the positional ODE defines the normalised clamped positional-flow density $\log q_\theta(x | c)$. (b) A universal neural potential assigns an energy value $E_k = E(c, x_k)$ to each of K displaced geometries of a single parent; only the values enter the objective. (c) $\mathcal{L}_{\text{value}}$ vanishes exactly when $\log q_\theta$ is affine in $-E_k/kT$ with unit slope and an arbitrary offset b , so the partition function cancels and is never estimated. (d) Groups A and B are two disconnected regions of the *same* $q_\theta(\cdot | c)$: each carries its own free offset, and the mass they receive relative to each other is unidentified (Appendix A.6).

Figure 2 (matched six-cell grid). Bars are means over independently trained seeds, whiskers the standard error over seeds, and every completed seed is drawn as an open symbol, because at two to five seeds the raw points are more informative than the interval; the grey band spans the observed seed range of the three no-physics cells A, B and C. Panel (b) reads the same seed means as differences Δr , with the propagated standard error over seeds $\sqrt{\text{sem}_a^2 + \text{sem}_b^2}$, which is a descriptive interval and not a confidence interval; at two to five seeds per arm these are effect sizes with a scale attached. The six differences are tabulated in Appendix B.9, and the D and F means remain conditional on completion, so their unconditional magnitudes are not identified.

Figure 3 (ordering and retrieval). Panel (a) plots the per-parent Pearson correlation averaged over parents, with open markers for seeds, the rule at the arm mean and the whisker the standard error over seeds; the bracket is the primary contrast. Panel (b) is the distribution of that correlation over the 93 parents, each averaged over its arm’s seeds. Panel (c) is retrieval: how often ranking a parent’s geometries by $\log q_\theta$ places the lowest-energy one first, or in the top three, with the chance rates for the $K = 8$ displaced geometries drawn as reference lines and the teacher-evaluator

agreement reference, which is ensemble-specific and not a ceiling, shown for context. Panel (d) sets the primary Gaussian-displacement contrast beside the fixed-RMSD family, in which the within-group displacement magnitude is held constant; because the two families are different perturbation populations, the comparable quantities are the two gaps and not the two levels, which the panel says on its axis. The fixed-RMSD side is drawn from arm means at four seeds per arm and carries no seed markers; the Gaussian side carries its raw seed points.

Figure 4 (ordering against calibration). Open markers are seeds; the filled marker or rule is the arm mean over seeds and the bar its standard error. Panel (a) puts ordering against raw calibration, where a constant log-density scores $\text{NRV} = 1$ exactly and the Boltzmann-consistent ideal is ($r = 1$, $\text{NRV} = 0$). Panel (b) shows the same seeds before and after rescaling each parent’s log-density by its own optimal factor; plotting $\text{NRV}_{\min}$, the mean over parents of the per-parent $1 - r_m^2$, as its own per-seed aggregate rather than as a $1 - r^2$ curve through the aggregate point of panel (a) is required because $\mathbb{E}_m[1 - r_m^2] \neq 1 - (\mathbb{E}_m r_m)^2$; the generator asserts the per-parent identity at build time. Panel (c) reads the same miss as a temperature. The orange marker in panels (a) and (b) is the teacher-evaluator agreement reference; its NRV , $\text{NRV}_{\min}$ and kT_{eff} were derived from the reference run’s per-parent table under the same aggregation conventions as the model arms.

B.19 REPRODUCTION

Environments. The pipeline runs in two Python environments, kept separate because the interatomic-potential stack and the graph-neural-network stack require incompatible versions of the deep-learning framework. The generator, its training loop and all density evaluations live in the first; dataset preprocessing and every teacher-energy evaluation live in the second. No step requires both simultaneously: all teacher labels used for training are pre-computed, and the two-stage evaluation communicates through a JSON file. Unit tests cover the score read-out against a finite-difference reference, the exact atom-wise divergence against the Hutchinson estimator, and the within-group variance loss.

Order of operations. The commands below reproduce the matched six-cell grid of Appendix B.9 from the public corpus, together with every number computed from it; each is a script in the accompanying archive, and the configuration file named on each training line is the only thing that differs between arms. They do *not* reproduce every training run in the paper: the earlier checkpoint family behind the primary endpoint was trained under a different shard list and different value-term settings, and its route is stated separately below.

1. *Preprocess* (second environment): `preprocess_omol25.py --split 4M --max-atoms 200`, then `write_omol25_valencies.py`, which emits the permissive element-valency file the sample analyser requires at construction time.
2. *Build the perturbation shards* (second environment; needed only when $\lambda_E > 0$): `precompute_energy_perturbations.py --split train --n-parents 30000 --k 5 --sigmas 0.03,0.06,0.10,0.20,0.40`. The y -randomisation shard is derived from that file by permuting the geometry-to-energy assignment within each parent (`--scramble_within_parent`); the zero-energy shard by `--zero_energies`. On the matched-grid route every shard an arm reads is built from the training split (Appendix B.4); the additional validation-split shard, built with `--split val --n-parents 10000`, is read only by the historical route below.
3. *Train* (first environment): `run_train.py --config <cell>_s<seed>.yaml`, one job per cell and seed, differing only in the physics block and the seed. Matched-grid configurations read the training-split shard only, take $M = 8$ parents per value step and cap the value loss at 1500. Diverged runs are recorded, not re-rolled.
4. *Evaluate, stage one* (first environment): `eval_boltzmann_stage1.py --checkpoint <ckpt> --n-molecules 120 --n-perturb 8 --sigma 0.15 --n-hutchinson 2 --eval-seed 12345`, which emits the perturbed geometries and their $\log q_\theta$.
5. *Evaluate, stage two* (second environment): `eval_boltzmann_stage2.py --samples-json <stage1.json>`, which scores the identical geometries with GFN2-xTB and writes one per-record CSV per arm.

6. *Aggregate*: `recompute_disjoint_noreference.py` restricts to the 93 disjoint parents and drops the data geometry; `analyze_calibration.py --kT_eV 1.0 --drop_reference` produces Tables A5 and A6; `retrieval_utility.py` produces Table A7; `measure_oracle_ceiling.py` produces the teacher-evaluator agreement reference; re-running `analyze_calibration.py` on copies of the same per-record files whose energy column has been replaced by the teacher energy for the matching (parent, displacement) pair produces the second block of Table A8; and `evaluate_validity.py --n 100` produces Table A14.

The two routes, and which numbers each one produces. *Matched-grid route.* The sequence above — training-split shards only, $M = 8$, value-loss cap 1500 — reproduces the matched six-cell grid, and hence Figure 2 and Appendix B.9. *Primary-endpoint historical route.* The earlier checkpoint family behind Table A4 was trained under the archived historical configurations, whose shard list carries the 10 000-parent validation-split shard alongside the 30 000-parent training-split shard, with $M = 4$ parents per value step and a value-loss cap of 3000 (Appendices B.4 and B.10). Reproducing that endpoint therefore means training from those archived configurations rather than from the matched-grid ones, evaluating 120 parents at step four, and applying the disjoint-parent filter of step six to obtain the 93-parent endpoint. The shard difference is not incidental: it is the defect audited in Appendix B.4, and it is why the endpoint is defined on the disjoint subset at all.

We will release a single anonymised archive containing both sets of configuration files, the launcher scripts, the evaluation scripts, the historical checkpoints and the per-record evaluation outputs, so that a reader who does not retrain can verify either route’s reported numbers from the records directly.

C EXTENDED SETTING AND RELATED WORK

Section 2.1 carries the compact positioning: the three groups of prior work, what each supplies, and the intersection this paper studies. This appendix adds what did not fit — the formal setting in practitioner’s terms, and three points of discussion that the main text states only in passing.

Setting. The formal version of what follows, with the standing assumptions, is Appendix A.1; this paragraph gives the practitioner’s reading. A molecular state is the pair (c, x) : a composition c — operationally the element multiset with the total charge, which is what the pipeline threads to the teacher — and a centre-of-mass-free conformation $x \in \mathbb{R}^{3N}$. Spin multiplicity is a standing restriction rather than a component of c : our teacher queries pass charge only and default to a singlet (Assumption 6), every evaluation reported here is on main-group organic parents at that default, and no radical, open-shell or transition-metal result is claimed. A teacher potential supplies $E(c, x)$ in eV and $F = -\nabla_x E$, hence $\pi(x | c) = e^{-E(c, x)/kT} / Z(c)$; neither $Z(c)$ nor $\sum_c Z(c)$ is ever formed. The teacher is the universal neural potential trained on OMol25 (Levine et al., 2025; Fu et al., 2025): 83 elements, DFT energies and forces, *no bond labels*. That is why the generator is bond-free rather than by preference — supervising bond orders would mean synthesising labels by rules least reliable exactly where broad element coverage is most valuable — and why bonds, where evaluation needs them, come post hoc (Kim & Kim, 2015). The backbone is FlowMol3 (Dunn & Koes, 2025), an SE(3)-equivariant conditional flow-matching model (Lipman et al., 2023; Liu et al., 2023; Albergo et al., 2023): $x_t = (1 - t)x_0 + tx_1$ with $x_0 \sim \mathcal{N}(0, \sigma^2 I)$ on the COM-free subspace, $\sigma = 1$, and $\mathcal{L}_{\text{FM}} = \mathbb{E} \|v_\theta(x_t, t) - (x_1 - x_0)\|^2$; types and charges ride its continuous-time Markov chain and are held at data values whenever an energy is queried.

What each group is judged on. The de novo generators of Section 2.1 (Hoogetboom et al., 2022; Xu et al., 2023; Vignac et al., 2023; Daigavane et al., 2024; Irwin et al., 2025; Dunn & Koes, 2025; Morehead et al., 2026) report validity, stability, novelty and distributional agreement with a reference corpus, and EBMol (Griesbacher et al., 2026) additionally learns a scalar energy, but that energy is fitted to data rather than supplied by a physical potential, so it is not an external target the density can be tested against. The conditional models (Jing et al., 2022; Axelrod & Gómez-Bombarelli, 2022; Moqvist et al., 2026) are judged by geometric agreement with a reference conformer ensemble for a given molecule. Neither family reports a density–energy check, which is the correct outcome for the questions they pose and not a defect; the metric of (4) is meaningful only once a tractable density and an external, transferable energy are both in hand.

The conditioning contrast, in full. Section 2.1 credits the log-variance divergence (Richter et al., 2020; Nüsken & Richter, 2021; Berner et al., 2024; Richter & Berner, 2024) and its off-policy estimation (Sendera et al., 2024) to the sampler literature. What differs here is the conditioning, not the divergence. Those methods train a sampler for one fixed target on one fixed state space; here the state space is itself generated, the target changes with every composition, and the divergence is summed over tens of thousands of parent groups whose normalisers are all distinct and all unknown. That summation is exactly what Theorem 2 analyses, and the price it charges — one free constant per connected component of the overlap graph — is the subject of Section 3. Mass-covering correction by annealed importance sampling (Midgley et al., 2023) is the closest existing remedy for the cross-group failure Corollary 2 formalises, on a state space where the components are known in advance. A different sense of “energy-based flow matching” also exists: Zhou et al. (2025) treat the reconstruction error of an idempotent flow map as an *internal, learned* energy, driving a predictor-refiner that iteratively improves conditional protein structures. Unlike IDFlow, we use values from an external universal potential to supervise and diagnose the likelihood of a bond-free de novo molecular flow.

Universal neural potentials, values and gradients. Potentials trained on large DFT corpora (Levine et al., 2025; Fu et al., 2025) are used in generative pipelines mostly downstream, to relax, rank or filter samples. The closest use of one as a training signal is Adjoint Sampling (Havens et al., 2025), which trains against an eSEN energy but samples conformers of a *given* molecular graph, whereas our generator receives nothing and never sees a bond label. Where a potential does supervise training the natural choice is its gradient, and force supervision works elsewhere (Guo et al., 2025; Hu, 2026); we therefore measure a gradient-supervision baseline alongside the value objective under a matched backbone and budget (Section 4.1), scoping its negative result to the construction we tested.